
Increasing Missingness to Reduce Bias: Richardson-SGD with Missing Data

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Abstract

1 Stochastic gradient methods are central to modern large-scale learning, but their use
2 with incomplete covariates remains delicate since imputation schemes generally
3 introduce systematic gradient biases, as shown for linear models. In this work, we
4 prove that all parametric models exhibit similar gradient bias for various imputation
5 procedures and characterize exactly the dependence on the missingness ratio vector
6 p , with $O(\|p\|)$ as the leading term. We exploit this analysis to propose a simple
7 debiasing procedure for stochastic gradient descent (SGD) with missing values
8 based on Richardson extrapolation, which leverages the exact expression of the
9 gradient bias. The key idea is to *deliberately add missingness*: from an already in-
10 complete observation, we generate a further-thinned version at a higher, controlled
11 missingness level, and combine the two resulting stochastic gradients to cancel the
12 leading bias term. We prove that one Richardson step reduces the gradient bias from
13 $O(\|p\|)$ to $O(\|p\|^2)$ under several missingness scenarios. Our proposed method is
14 computationally efficient, model-agnostic and applies to any parametric loss whose
15 stochastic gradient can be computed after imputation. Furthermore, when missing
16 indicators are independent, the population gradient bias is a multilinear polynomial
17 in p and depends only on population gradient errors induced by declaring a single
18 coordinate missing. In this case, our method generalizes to a multi-step Richardson
19 procedure which recursively cancels higher-order terms. Empirically, Richardson
20 debiasing improves optimization and estimation across several generalized linear
21 models and combines positively with widely used imputation procedures such as
22 MICE. These results suggest that, somewhat counter-intuitively, adding controlled
23 missingness on top of existing missing data can make stochastic learning from
24 incomplete data more accurate.

1 Introduction

26 Missing data are ubiquitous in modern machine learning. They may arise from database fusion,
27 sensor failure, non-response in surveys, and selective acquisition pipelines, to name only a few.
28 In his seminal paper, Rubin [30] formalized the missing-data framework and introduced the now-
29 standard taxonomy of three missingness regimes: *Missing Completely at Random* (MCAR), in which
30 missingness is independent of the data; *Missing at Random* (MAR), in which missingness depends
31 only on observed entries; and *Missing Not at Random* (MNAR), in which missingness can also
32 depend on the unobserved entries themselves.

33 **Framework.** In supervised learning with incomplete covariates, one typically distinguishes two goals:
34 estimating the parameters of a model despite the missing values, and producing a predictor with high
35 test accuracy. The two are aligned when the test set is fully observed—accurate parameter estimation
36 then leads to strong predictive performance—but they decouple when the test set itself contains
37 missing entries, in which case a separate prediction-time strategy is required [see e.g. 14, 35, 13]. We

focus on the first objective and assume that missing values appear only in the training set, while the test set is complete. Even in this setting, parameter identifiability is not guaranteed under arbitrary MNAR mechanisms [see, e.g., the examples and discussions in 29, 38, 20]. We therefore restrict our attention to MCAR and a generalization—scalable MAR—in which the conditional missingness probability depends on a known intensity function.

Handling missing data in parametric models. The simplest approach is *complete-case* analysis [26, 16], which discards every sample containing at least one missing entry. This is unbiased under MCAR but throws away samples at a rate that is exponential in the dimension. The next-simplest approach is *imputation*: missing entries are replaced by point estimates, after which any standard learning algorithm can be applied to the completed dataset. Constant imputation (zero or mean) is the most studied [12] and the easiest to analyze, but it injects a systematic bias even under MCAR. Multivariate Imputation by Chained Equations [MICE, 34] and nearest-neighbour or neural-network imputation schemes [33, 19] reduce this bias empirically but offer few formal estimation guarantees. A complementary line of work avoids imputation altogether by working with the joint distribution of inputs and mask. The Expectation–Maximization algorithm of Dempster et al. [9], refined for incomplete data by Ibrahim [10] and extended to logistic regression via Stochastic Approximation EM [11], fits a parametric model to the inputs and the predictor jointly. These algorithms require a known parametric family for the covariates and can be expensive due to the E-step. A first review of estimation procedures for linear regression with missing covariates was given in Little [15].

Related work - SGD with missing data. Gradient descent and its stochastic variants are the workhorse of large-scale learning, but they require a fully observed input to compute a gradient. The natural fix is to impute and then run SGD on the completed dataset; as observed by Jones [12], this leads to a biased gradient. Ayme et al. [2, 3] study the test-time predictive performance of SGD applied to zero-imputed data when data can be missing in both train and test set, relating the imputation bias to a ridge regularization effect and leveraging the implicit bias of SGD [31] towards low-norm solutions to derive convergence rates [see also 7, 36, for missing data in high-dimensional linear models]. Another line of work focuses on parameter estimation or equivalently on test-time performance when the test set is assumed to contain complete data. Loh and Wainwright [17] characterize the exact bias induced by zero imputation in linear models and use this characterization to obtain the first parameter-estimation rates for sparse high-dimensional linear regression under MCAR. Building on their analysis, Needell [23] design a stochastic gradient algorithm that is better suited to large-scale data, and Sportisse et al. [32] establish that averaged debiased SGD attains the optimal one-pass rate for linear regression.

Contributions. We propose and analyze a debiasing procedure for stochastic gradients computed from imputed data, which can be applied to any parametric model and which is valid for a large class of imputation procedures. Our proposed method applies Richardson extrapolation [28] to the missingness scale p , yielding a model-agnostic correction that can be combined with a broad class of imputation rules. While Richardson extrapolation has been used in machine learning to remove leading-order biases in other contexts [5], to the best of our knowledge, this is the first application to the missingness scale of a stochastic gradient. Our contributions are as follows.

Gradient-bias structure. Under several MCAR and MAR settings described below, we establish the exact expression of the population bias of a stochastic gradient computed on imputed data. In doing so, we generalize the expression obtained by Sportisse et al. [32] for linear regression with zero-imputed data and independent MCAR missingness to any parametric model, a large class of imputation procedures, and a broad class of missingness scenarios (non-independent MCAR and MAR). As a consequence, we show that the gradient bias is $O(\|p\|)$, where $p = (p_1, \dots, p_d)$ with p_j being the probability that the j th component is missing. When the mask components are independent, we prove that the order of the remaining terms is $O(\|p\|^2)$. Our bias decomposition holds for generic imputation procedures: better imputation may shrink the constants in $O(\|p\|)$ but cannot generally remove the leading $O(\|p\|)$ bias (Section 3).

Richardson-SGD. We introduce a thinning construction that, from a sample with mask at scale p , generates a further-thinned mask at scale Cp , for some well-chosen $C > 1$, using one extra Bernoulli draw per observed entry. A Richardson combination of the two gradients (computed on an imputed dataset at scales Cp) reduces the bias from $O(\|p\|)$ to $O(\|p\|^2)$ under some MCAR and MAR settings with independent masking components (Section 4). We also introduce a multi-step Richardson-based procedure which cancels higher-order terms, with exact cancellation for d_{miss} steps, where d_{miss} is the number of covariates subject to missingness.

95 *Theory for one-pass SGD.* Our bias expansion plugs directly into classical biased-SGD proofs. For
 96 one-pass (one-epoch) SGD over n samples, and given a smooth and strongly convex loss, Richardson-
 97 SGD attains $\mathbb{E}\|w_n - w^*\|^2 = O(\|p\|^4) + O(1/n)$, against $O(\|p\|^2) + O(1/n)$ for plain imputed
 98 SGD. Multi-step Richardson reduces the missingness term further at the price of increased variance.
 99 (Section 5).

100 *Experiments.* We validate the theory on synthetic and real datasets. For a variety of generalized
 101 linear models, Richardson-SGD improves over plain imputation under several MCAR and MAR
 102 mechanisms and combines positively with MICE, Random-Forest MICE, and k -NN imputation
 103 (Section 6).

104 2 Setting

105 **Random covariates and notation.** Random variables are written in uppercase (X, Y, M) ; their
 106 realizations are written in the corresponding lowercase (x, y, m) . For an integer $d \geq 1$ we set
 107 $[d] := \{1, \dots, d\}$. For any $S \subseteq [d]$, we write $S^c := [d] \setminus S$, and for a vector $v \in \mathbb{R}^d$ we let
 108 $v^{(S)} \in \mathbb{R}^{|S|}$ denote the subvector indexed by S . Throughout the paper, $\|\cdot\|$ is the Euclidean norm
 109 and $\|\cdot\|_\infty$ the supremum norm.

110 **Supervised learning and SGD.** We consider a supervised learning setting with random covariates
 111 $X \in \mathcal{X} \subseteq \mathbb{R}^d$, response $Y \in \mathcal{Y}$, parameter $w \in W \subseteq \mathbb{R}^q$, and a differentiable loss $\ell : W \times \mathcal{X} \times \mathcal{Y} \rightarrow$
 112 $\mathbb{R}$. We aim at minimizing the population risk $L(w) := \mathbb{E}[\ell(w; X, Y)]$. The complete-data single-
 113 sample gradient is $g(w; x, y) := \nabla_w \ell(w; x, y)$. Assuming differentiation and expectation commute,
 114 we have $\nabla L(w) = \mathbb{E}[g(w; X, Y)]$. Because ∇L has no closed form in general, we use stochastic
 115 gradient descent (SGD), whose updates are given by

$$w_{k+1} = w_k - \eta_k \hat{g}_k(w_k), \quad (1)$$

116 where $\eta_k > 0$ is the step-size and $\hat{g}_k(w_k)$ is a stochastic estimator of $\nabla L(w_k)$ computed at iteration
 117 k from a sample or minibatch. When the sample is complete, $g(w; X, Y)$ is unbiased for $\nabla L(w)$.

118 **Missing covariates and imputation.** For each training sample, the learner observes a realisation of
 119 (X^{obs}, Y, M) , where $M \in \{0, 1\}^d$ is a missingness mask and, for all $j \in [d]$, $X_j^{\text{obs}} = X_j$ if $M_j = 0$
 120 and $X_j^{\text{obs}} = \text{NA}$ if $M_j = 1$. We write $p_j := \mathbb{P}(M_j = 1)$ for the marginal missingness probability
 121 of feature j and $p := (p_1, \dots, p_d)$ for the missingness vector. Missing entries are filled in by an
 122 imputation rule $\mathcal{I}$ that produces an imputed covariate vector

$$\tilde{X} := \mathcal{I}(X^{\text{obs}}, M, \xi), \quad (2)$$

123 where $\xi \perp M \mid (X, Y)$ collects auxiliary randomness used by $\mathcal{I}$. We focus on *data-independent*
 124 *imputation rules* $\mathcal{I}$ that impute each observation independently of the others. This assumption makes
 125 our analysis tractable by enabling a decomposition at the sample level. Standard imputations (mean,
 126 iterative) can be slightly modified to fall into this setting by training $\mathcal{I}$ on an auxiliary dataset. The
 127 imputed stochastic gradient available to the learner is $\hat{g}(w) := g(w; \tilde{X}, Y)$, which is, in general, a
 128 *biased* estimator of $\nabla L(w)$, with bias

$$\mathcal{B}(w, p) := \mathbb{E}[\hat{g}(w)] - \nabla L(w), \quad (3)$$

129 the expectation being over (X, Y) , the mask M , and the imputation randomness ξ .

130 **Missingness mechanisms.** We follow the taxonomy of Rubin [30]: the mask is *Missing Completely*
 131 *at Random* (MCAR) when $M \perp (X, Y)$, and *Missing at Random* (MAR) when, conditionally on
 132 the observed entries, M is independent of the missing entries. Throughout the paper, we let $\mathcal{O} \subseteq [d]$
 133 (possibly empty) be the set of indices of variables that are *always-observed*. We let $V := X^{(\mathcal{O})}$ be
 134 the vector of *always-observed variables*. To enable a tractable analysis, we focus on two concrete
 135 mechanisms, which depend on the probability vector p , assumed to be known.

136 **Heterogeneous MCAR (hMCAR).** M is independent of (X, Y) and $\mathbb{P}(M_j = 1) = p_j$.

137 **Scalable MAR (sMAR).** $\{M_j\}_{j \in \mathcal{O}^c} \perp (X^{(\mathcal{O}^c)}, Y) \mid V$, and for every $j \in \mathcal{O}^c$,

$$\mathbb{P}(M_j = 1 \mid V) = p_j q_j(V),$$

for known intensity functions $q_j : \mathbb{R}^{|\mathcal{O}|} \rightarrow [0, p_j^{-1}]$ with $\mathbb{E}[q_j(V)] = 1$.

It is known that MAR settings contain scenarios of different difficulties [22], some of which being close to MNAR settings [21], for which identifiability does not always hold [see, e.g., 29, 38, 20]. Thus, we restrict the MAR settings we consider via the sMAR assumption. Note that the condition $\mathbb{E}[q_j(V)] = 1$ in sMAR is necessary to ensure that $p_j = \mathbb{P}(M_j = 1)$. A concrete example of sMAR is a logistic missingness mechanism, as commonly used in simulation studies of missing covariates [e.g. 18, 37]. We say a mask is *independent hMCAR* (resp. *independent sMAR*) if it is hMCAR (resp. sMAR) and the $\{M_j\}_{j \in \mathcal{O}^c}$ are mutually independent (resp. conditionally on V).

Our objective remains the complete-data risk $L(w)$ and its minimizer w^* ; missingness and imputation only affect the stochastic gradients used to optimize it. Our goal is to replace the imputed gradient $\hat{g}$ in (1) by a corrected gradient $\hat{g}^R$, computed from the same observation plus a small amount of controlled additional thinning, so as to cancel or shrink the gradient bias (3).

3 First-order structure of the missingness bias

Before designing a debiasing procedure, we describe the structure of the imputation-induced gradient bias as a function of the missingness scale p . The key observation is that, regardless of the loss and the imputation rule, the bias admits a clean expansion whose leading term is *linear* in p and whose coefficients are population gradient gaps that do not depend on p . This expansion will be the structural fact that Richardson extrapolation later exploits.

Proposition 1 (First-order structure of the missingness bias). *Consider any data-independent imputation defined in (2) and assume hMCAR or sMAR holds. Then the population gradient bias (3) can be decomposed as*

$$\mathcal{B}(w, p) = \mathcal{A}(w)p + \mathcal{R}(w, p), \quad (4)$$

with $\mathcal{A}(w) \in \mathbb{R}^{q \times d}$ independent of p . Letting $a_j(V) = 1$ for hMCAR and $a_j(V) = q_j(V)$ for sMAR, the j -th column of $\mathcal{A}(w)$ is the population gradient gap obtained by declaring coordinate j missing:

$$\mathcal{A}_{\cdot j}(w) = \mathbb{E} \left[a_j(V) \left\{ G_{\{j\}}(w; X, Y, \xi) - \nabla L(w) \right\} \right]. \quad (5)$$

The remainder $\mathcal{R}(w, p)$ contains the co-missingness contributions, namely the terms involving simultaneous missingness of two or more coordinates. The exact expression of $\mathcal{R}(w, p)$ is given in Appendix B. The proof is based on a discrete-difference expansion over missingness patterns and separates the contribution of each joint missingness pattern $S \subseteq [d]$.

In full generality, the remainder is at most linear in p , while it is $o(\|p\|)$ in most scenarios. Indeed, strong dependence among mask components can make co-missingness terms contribute at first order. For instance, this may occur when two coordinates are perfectly negatively associated, so that $\mathbb{P}(M_j = 1 \mid M_k = 1) = 0$. The following corollary identifies a key regime motivating Richardson extrapolation.

Corollary 1. *Under the assumptions of Proposition 1, suppose in addition that the missingness indicators $\{M_j\}_{j \in \mathcal{O}^c}$ are conditionally independent given V . Then*

$$\|\mathcal{R}(w, p)\| = O(\|p\|^2), \quad \text{and therefore} \quad \|\mathcal{B}(w, p) - \mathcal{A}(w)p\| = O(\|p\|^2). \quad (6)$$

Under independent hMCAR (resp. independent sMAR), the full bias is a *multilinear polynomial* in p ,

$$\mathcal{B}(w, p) = \sum_{\emptyset \neq S \subseteq [d]} \mu_S(w) \left(\prod_{j \in S} p_j \right), \quad \mu_S(w) := \mathbb{E} \left[\left(\prod_{j \in S} a_j(V) \right) \Delta_S G_{\emptyset}(w; X, Y, \xi) \right], \quad (7)$$

where $T_j G_{\emptyset} := G_{\{j\}}$ declares coordinate j missing and $\Delta_S G_{\emptyset} := \prod_{j \in S} (T_j - I) G_{\emptyset}$. The coefficient $\mu_S(w)$ aggregates the effect of $|S|$ -fold co-missingness. Equation (7) is the structural fact that drives both first and higher-order Richardson cancellation.

The decomposition has three implications. (i) *The leading bias is linear in p* : under conditional independence, the remainder is $O(\|p\|^2)$, so the first-order behavior is fully captured by $\mathcal{A}(w)p$. (ii) *The leading operator is an average gradient gap*: the column $\mathcal{A}_{\cdot j}(w)$ vanishes whenever coordinate j is always observed or is perfectly recovered by $\mathcal{I}$. (iii) *Imputation reduces constants, not the leading*

180 *order*: the expansion holds for any data-independent imputation, and a better imputation rule only
 181 shrinks the entries of $\mathcal{A}_j(w)$ without changing the order of $\mathcal{B}(w, p)$ in p .

182 These three points together suggest a clear strategy. Imputation alone cannot remove the leading
 183 $O(\|p\|)$ scaling, except if it fully recovers the covariate. Improving the imputation only refines the
 184 constants $\mathcal{A}_j(w)$. To eliminate the leading order, we propose to act *on p itself*—that is, evaluate
 185 the imputed gradient at two different missingness scales and combine the results so that the linear
 186 contribution cancels. This is precisely what Richardson extrapolation achieves, and the construction
 187 we develop in the next section turns this idea into a practical SGD update.

188 4 Richardson-SGD

189 **Richardson extrapolation in a nutshell.** Richardson extrapolation [28] cancels the leading term of
 190 an asymptotic expansion. If $T(p) = T_0 + pT_1 + p^2T_2 + o(p^2)$ as $p \rightarrow 0$ and $C > 1$, the combination

$$T_C^R(p) := \frac{C T(p) - T(Cp)}{C - 1} = T_0 - Cp^2T_2 + o(p^2) \quad (8)$$

191 eliminates the linear term. With $k + 1$ scales $1 = C_0 < C_1 < \dots < C_k$ and a Vandermonde weight
 192 vector, the first k orders are cancelled simultaneously [24].

193 At first sight, applying (8) to the missingness bias would require evaluating the imputed gradient at
 194 *two* missingness scales p and Cp on the same observation. The learner, however, only observes a
 195 single mask $M^{(p)}$ at scale p . We resolve this with a single extra Bernoulli draw per observed entry:
 196 from a sample at scale p , we *further thin* it to obtain a mask whose conditional law given X is exactly
 197 that of an independent draw at scale Cp . No new observation is required. We employ this additional
 198 mask to propose a Richardson-corrected gradient, used in lieu of the standard gradient in a SGD
 199 procedure.

200 **Further-thinned mask.** Fix $C > 1$ such that, for all $j \in \mathcal{O}^c$, $Cp_j a_j(V) \leq 1$ almost surely.
 201 Conditional on $(X, M^{(p)})$, draw independent thinning bits r_j with $r_j = 0$ for $j \in \mathcal{O}$ and, for $j \in \mathcal{O}^c$,

$$r_j \mid (X, M^{(p)}) \sim \text{Bernoulli}\left(\frac{1 - Cp_j a_j(V)}{1 - p_j a_j(V)}\right), \text{ and let } M_j^{(Cp)} := 1 - (1 - M_j^{(p)}) r_j. \quad (9)$$

202 All entries missing under $M^{(p)}$ stay missing under $M^{(Cp)}$; an observed entry is hidden under $M^{(Cp)}$
 203 exactly when $r_j = 0$. A short calculation (Appendix C) gives $\mathbb{P}(M_j^{(Cp)} = 1 \mid X) = Cp_j a_j(X)$, so
 204 $M^{(Cp)}$ has the same distribution as that of the original mask but at scale Cp .

205 **Richardson-corrected gradient.** Equipped with the further-thinned mask, we can apply (8) to the
 206 imputed gradient. Crucially, we must *not* impute the same observation at two different missingness
 207 levels, since we need common missing values between the two scales to be identical (see Appendix L
 208 for further explanation and a numerical illustration). We impute once on the more thinned sample at
 209 scale Cp , then *restore* the artificially hidden entries to recover the imputation at scale p :

$$\tilde{X}^{(Cp)} := \mathcal{I}(X^{\text{obs}}, M^{(Cp)}, \xi), \quad \tilde{X}_{Cp,j}^{(p)} := \begin{cases} X_j, & M_j^{(p)} = 0, \\ \tilde{X}_j^{(Cp)}, & M_j^{(p)} = 1, \end{cases} \quad \text{for all } j \in [d].$$

210 Set $\hat{g}^{(p)}(w) := g(w; \tilde{X}^{(p)}, Y)$ and $\hat{g}^{(Cp)}(w) := g(w; \tilde{X}^{(Cp)}, Y)$. The *Richardson-corrected gradient*
 211 is

$$\hat{g}_C^R(w) := \frac{C \hat{g}^{(p)}(w) - \hat{g}^{(Cp)}(w)}{C - 1}. \quad (10)$$

212 **Richardson-SGD** plugs $\hat{g}_C^R$ into the SGD update (1). For each sampled observation $(x_i^{\text{obs}}, m_i, y_i)$ at
 213 iteration k :

- 214 1. *Original masked sample.* Read off the mask m_i at scale p .
- 215 2. *Further-thinned sample.* Draw r as in (9). For all $j \in [d]$, $m_{ij}^{(Cp)} \leftarrow 1 - (1 - m_{ij}^{(p)}) r_j$.
- 216 3. *One imputation.* Compute $\tilde{x}_i^{(Cp)} \leftarrow \mathcal{I}(x_i^{\text{obs}}, m_i^{(Cp)}, \xi_i)$, then obtain $\tilde{x}_i^{(p)}$ by overwriting the
 217 entries hidden by r with their true values from x_i .

- 218 4. *Gradient estimates.* Evaluate $\hat{g}_i^{(p)} := g(w_k; \tilde{x}_i^{(p)}, y_i)$ and $\hat{g}_i^{(Cp)} := g(w_k; \tilde{x}_i^{(Cp)}, y_i)$.
 219 5. *Richardson correction & SGD update.* Form $\hat{g}_k^R \leftarrow (C \hat{g}_i^{(p)} - \hat{g}_i^{(Cp)}) / (C - 1)$ and update
 220 $w_{k+1} \leftarrow w_k - \eta_k \hat{g}_k^R$ (averaged across a minibatch when $b > 1$).

221 The procedure is a thin wrapper around any imputation-based SGD pipeline: one extra Bernoulli
 222 draw per observed entry and one extra gradient evaluation per sample.

223 5 Theory of Richardson-SGD

224 We now state the theoretical guarantees of Richardson-SGD. The analysis shows that Richardson
 225 corrections successively cancel the terms in the bias expansion, while controlling the associated
 226 variance inflation and the error from estimating the missingness mechanism. Combining these bounds
 227 with a classical biased-SGD argument yields a convergence rate. Throughout, the result applies to
 228 *one-pass (one-epoch) SGD*, as in Sportisse et al. [32] for linear regression: each sample is visited
 229 once, and the bias expansion from Section 3 feeds directly into standard biased-SGD arguments.
 230 Multi-epoch behavior is outside the scope of the theory and is examined empirically in Section 6.

231 5.1 First-order bias cancellation

232 **Proposition 2** (First-order debiasing). *Assume hMCAR or sMAR. Then*

$$\| \mathbb{E}[\hat{g}_C^R(w)] - \nabla L(w) \| = o(\|p\|), \quad \text{when } \|p\| \rightarrow 0. \quad (11)$$

233 *For independent hMCAR or independent sMAR, the right-hand side term turns into $O(\|p\|^2)$.*

234 Proposition 2 shows that the debiasing challenge can be met by a deliberately counterintuitive
 235 operation: we decrease bias by adding missing values. While the plain imputed gradient has bias
 236 of order $\|p\|$, the Richardson-corrected gradient constructed from the original and further-thinned
 237 masks cancels this leading term and leaves only an $O(\|p\|^2)$ bias under independent hMCAR or
 238 independent sMAR. This gain is uniform in the loss and the imputation rule, and requires only one
 239 additional Bernoulli draw and one additional gradient evaluation per sample (proof in Appendix C).

240 5.2 Higher-order Richardson-SGD under independent masks

241 When the missing indicators are conditionally independent given V , Section 3 showed that the
 242 gradient bias is, in fact, a multilinear polynomial in p . Since Richardson extrapolation is itself linear
 243 in the underlying expansion, one can cancel further orders by combining estimators at more than two
 244 missingness scales. Iterating the thinning construction with $k + 1$ scales $1 = C_0 < C_1 < \dots < C_k$
 245 (cascaded via (9) with $C \leftarrow C_\ell / C_{\ell-1}$) and Vandermonde weights $\alpha \in \mathbb{R}^{k+1}$ yields the k -th order
 246 Richardson estimator $\hat{g}^{[k]}(w) := \sum_{\ell=0}^k \alpha_\ell \hat{g}^{(C_\ell p)}(w)$.

247 **Corollary 2** (Higher-order cancellation). *Under independent hMCAR or independent sMAR,*
 248 $\| \mathbb{E}[\hat{g}^{[k]}(w)] - \nabla L(w) \| = O(\|p\|^{k+1})$ *as $\|p\| \rightarrow 0$. Furthermore, with $d_{\text{miss}} := \text{Card}(\{j : p_j > 0\})$,*
 249 *the d_{miss} -th order estimator cancels the bias exactly: $\mathbb{E}[\hat{g}^{[d_{\text{miss}}]}(w)] = \nabla L(w)$.*

250 For linear regression with squared loss, the bias is a polynomial of degree at most 2 in p (Appendix E),
 251 so the two-step Richardson-SGD produces an *exact* debiasing under both hMCAR and sMAR. This
 252 matches the closed-form correction mechanism of Sportisse et al. [32] as a special case and extends it
 253 to sMAR, where no closed form is available. More generally, Corollary 2 suggests that higher-order
 254 Richardson-SGD should be most useful when only a few coordinates are subject to missingness (d_{miss}
 255 small) so that the corresponding polynomial degree is low, or that the highest polynomial degree in
 256 the bias is low, as for linear regression (see Appendix D). Figure 1 illustrates this phenomenon in
 257 synthetic linear and logistic regressions.

258 5.3 Variance inflation

259 The previous two subsections highlight how Richardson reduces bias. As is standard in extrapolation
 260 methods, this comes at a price: the corrected gradient is a difference of two estimators evaluated at
 261 different missingness levels, which inflates its variance. Quantifying this inflation is essential, since

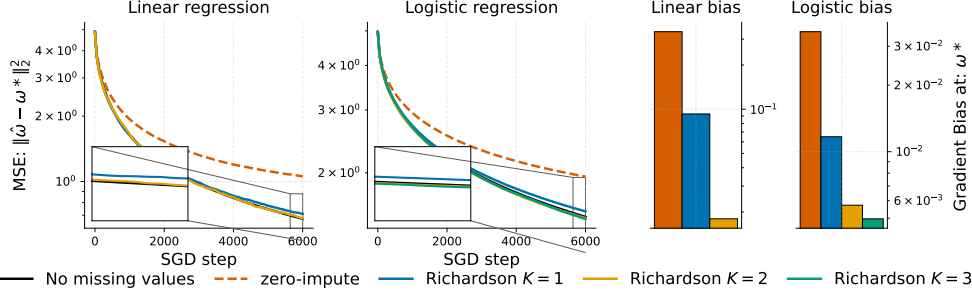


Figure 1: Multi-order Richardson correction in 4-covariate linear and logistic regression under hMCAR with $p = (0.10, 0.15, 0.08, 0.12)$, and bias at w^* , $\|\mathbb{E}[\hat{g}(w^*)] - \nabla L(w^*)\|$. As predicted by Corollary 2, each Richardson level removes one further order of bias: first-order Richardson clearly improves over standard SGD on zero-imputed data, second-order matches the complete-data trajectory in linear regression (curves overlap), and third-order yields an additional gain in the logistic case for the bias.

the convergence rate of SGD depends on *both* the bias and the variance of the stochastic gradient. For the first-order estimator,

$$\text{Var}[\hat{g}_C^R(w)] \leq \frac{C^2 \text{Var}[\hat{g}^{(p)}(w)] + \text{Var}[\hat{g}^{(Cp)}(w)]}{(C-1)^2}, \quad (12)$$

with a larger C favoring variance control since $x : (1, +\infty) \mapsto x^2/(x-1)^2$ is decreasing. For the k -th order estimator, variance inflates by a factor that grows with k , and requires k missingness upscales, which limits k when some p_j are large. We therefore use first-order Richardson by default and reserve higher-order constructions for small d_{miss} or for losses with low maximum polynomial degree, as linear regression, which is of degree 2 (see Appendix D).

5.4 Richardson-SGD with estimated missingness parameters

So far we have assumed that the quantities (p, q) driving the missing mechanism are known. In practice, p_j is estimated by the empirical missingness frequency on coordinate j , while $q_j(V)$ is fitted by a probabilistic model with input V . We now quantify how the resulting estimation errors propagate into the Richardson bias. Using the identifiability convention $\mathbb{E}[q_j(V)] = 1$, let $\lambda_j(V) := p_j q_j(V)$ and $\hat{\lambda}_j(V) := \hat{p}_j \hat{q}_j(V)$. The plug-in thinning rule replaces (9) by $r_j \sim \text{Bernoulli}((1 - C\hat{\lambda}_j(V))/(1 - \hat{\lambda}_j(V)))$ and yields the plug-in Richardson gradient $\hat{g}_{C, \hat{\lambda}}^R$.

Proposition 3 (Plug-in Richardson). Assume hMCAR or sMAR with $\lambda_j(V), \hat{\lambda}_j(V) \leq \rho < 1$ and $\|G_S(w; X, Y, \xi) - \nabla L(w)\|_{L^2} \leq G_*$ for every $S \subseteq [d]$. If $\|\hat{p} - p\|_\infty \leq \delta_p$ and $\max_j \sup_v |\hat{q}_j(v) - q_j(v)| \leq \delta_q$, then

$$\|\mathbb{E}[\hat{g}_{C, \hat{\lambda}}^R(w)] - \nabla L(w)\| = O(\|p\|^2 + \delta_p + \|p\|_\infty \delta_q + \delta_p \delta_q). \quad (13)$$

Under hMCAR ($q_j \equiv 1, \delta_q = 0$), this collapses to $O(\|p\|^2 + \delta_p)$.

The leading $\mathcal{A}(w)p$ contribution is cancelled regardless of plug-in errors, up to an additive $O(\delta_p + \|p\|_\infty \delta_q)$ penalty (proof in Appendix C.4). When δ_p, δ_q shrink fast enough, the $O(\|p\|^2)$ term dominates and the exact-mechanism guarantee is recovered. This behavior further motivates using the first-order Richardson SGD scheme, while higher order might not be conclusive in the plug-in setting. Appendix H reports an empirical sensitivity study.

5.5 One-pass SGD convergence

We have now controlled both the bias of the Richardson-corrected gradient, through Proposition 2 and Corollary 2, and its variance, through (12), including under plug-in mechanisms (Proposition 3). It remains to translate these gradient-level guarantees into a convergence rate for the SGD iterates, which is ultimately the quantity of interest. Note that biased SGD schemes have been extensively studied in the literature [see, e.g. 1, 8]. To illustrate the resulting bias improvement of Richardson-SGD compared to plain imputation, we give a result under classic regularity conditions on the loss function.

293 **Corollary 3** (One-pass Richardson-SGD). Assume L is α -strongly convex and β -smooth, the per-
 294 sample stochastic gradients are bounded in L^2 . Under independent hMCAR or independent sMAR,
 295 after one pass on n i.i.d. samples with $\eta_k = \frac{c}{k+\gamma}$, $c > \frac{1}{\alpha}$, $\gamma \geq \frac{6c\beta^2}{\alpha}$, we obtain

$$\mathbb{E}\|w_n - w^\star\|^2 = \begin{cases} O(\|p\|^2) + O(1/n) & (\text{plain imputed SGD}), \\ O(\|p\|^4) + O(1/n) & (\text{Richardson-SGD, first order}), \end{cases} \quad (14)$$

296 and the same convergence orders hold for the excess test loss $\mathbb{E}[L(\bar{w}_n) - L(w^\star)]$. With k -step
 297 Richardson-SGD, the missingness floor becomes $O(\|p\|^{2(k+1)})$. Thus, for sufficiently large k , the
 298 missingness contribution is dominated by the statistical floor $O(1/n)$.

299 Two implications of Corollary 3 are worth highlighting. First, when $\|p\| \ll 1$, one-step Richardson-
 300 SGD improves the bias floor of plain imputed SGD from $O(\|p\|^2)$ to $O(\|p\|^4)$. Thus, the missingness-
 301 induced contribution is reduced by two orders of magnitude in $\|p\|$, while keeping essentially the
 302 same per-iteration cost. Second, multi-step Richardson-SGD can, in principle, reduce the missingness
 303 term down to the statistical noise level $O(1/n)$. This comes at the price of variance inflation: the
 304 bound in (12) compounds across Richardson levels and may become prohibitive when d_{miss} is large
 305 or when the loss has heavy stochastic gradients. Consequently, multi-step Richardson-SGD is most
 306 appealing when d_{miss} is small, or in settings such as linear regression where order 2 already suffices.

307 **Scope of the theory.** We emphasize that Corollary 3 is a one-pass guarantee, in line with the regime
 308 studied by Sportisse et al. [32]. The multi-epoch behavior is not covered by our analysis: when
 309 iterates revisit the same observations, the gradient noise due to missing values across iterations are no
 310 longer independent. The experiments of Section 6 suggest, however, that Richardson-SGD remains
 311 effective in multi-epoch training, and we view a formal multi-epoch analysis as an interesting question
 312 for future work.

313 6 Experiments

314 We empirically study Richardson-SGD on synthetic and real datasets available in `scikit-learn`
 315 [25]. Throughout, missing entries are introduced *ex post* into otherwise complete datasets according
 316 to the mechanism specified in each subsection, either hMCAR or sMAR, so that the ground truth $w^\star$
 317 is known, or can be estimated by multi-pass training with L-BFGS-B, and the quantities p and q_j are
 318 also known. Unless stated otherwise, the average missingness is fixed at $\bar{p} = 0.2$. To keep the main
 319 text concise, we report only logistic regression here; analogous experiments for other datasets and
 320 models, including linear and Poisson GLMs, together with implementation details, are deferred to
 321 Appendix I.

322 **Empirical takeaway.** Across datasets, models, missingness mechanisms, and imputation rules,
 323 Richardson-SGD behaves as a generic debiasing layer rather than a model-specific correction. It
 324 improves imputation-based SGD using only controlled thinning and one additional gradient evaluation,
 325 and remains effective when the missingness mechanism is estimated or partially misspecified. In
 326 short, the method is simple, fast, model-agnostic, and theoretically grounded, making it a natural
 327 add-on for learning with missing covariates.

328 6.1 Richardson with imputation on logistic regression

329 This experiment tests the central practical claim of the paper: *Richardson extrapolation can be*
 330 *combined effectively with standard imputation methods*. We run logistic regression under hMCAR
 331 missingness, comparing SGD applied on the most standard imputations (namely MICE, MICE with
 332 random-forest base learners, and k -nearest-neighbor imputations) used in conjunction with SGD, and
 333 the Richardson-SGD counterparts (applied to the same imputation procedures). Across missingness
 334 levels and datasets, Richardson consistently acts as a complementary debiasing layer: the imputer
 335 reduces the initial missingness bias, while Richardson further reduces the residual gradient bias, with
 336 the largest gains obtained when the underlying imputer is already accurate.

337 **Additional experiments.** Appendix I extends the numerical study beyond logistic regression to
 338 several other Generalized Linear Models (linear and Poisson), datasets, imputation rules, and missing-
 339 ness mechanisms. Across settings, Richardson consistently improves the considered imputations and
 340 remains effective beyond the one-pass regime. The gains are largest in the first epoch, matching the

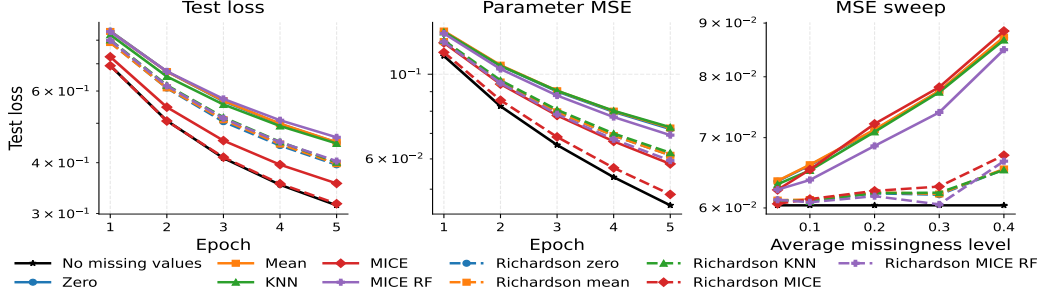


Figure 2: **Covertypes missingness sweep.** Test loss and parameter mean-squared error for logistic regression on Covertypes [6] under hMCAR missingness, as the average missingness level $\bar{p}$ varies. Richardson improves over each corresponding imputed SGD baseline across a broad range of $\bar{p}$, showing that the correction is not limited to the very small-missingness regime.

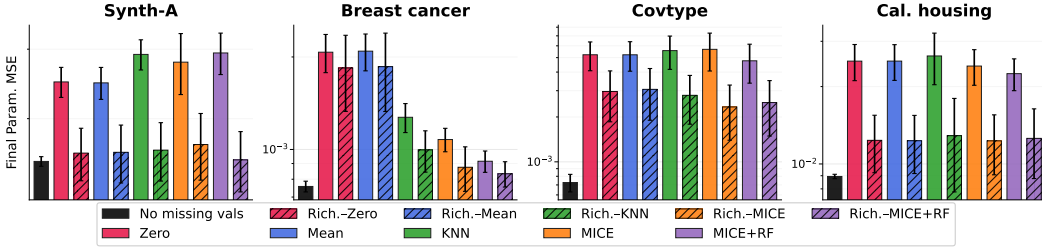


Figure 3: **Cross-dataset comparison.** Final parameter mean-squared error of the last SGD iterate for logistic regression under hMCAR missingness across multiple datasets. Each imputation-based SGD baseline is compared with its first-order Richardson-SGD counterpart. Richardson systematically lowers the final parameter error, with particularly clear gains on the Breast Cancer dataset [39], where stronger imputers lead to substantially smaller Richardson-corrected errors.

theory of Section 5.5. We also show robustness to estimated missingness values by replacing p and q with their estimates in Appendix H, and robustness to misspecification of the missingness mechanism by using Richardson-SGD under an assumed hMCAR mechanism while the true mechanism is sMAR in Appendix K.

7 Conclusion

We introduced Richardson-SGD, a simple debiasing method for stochastic gradient learning computed on imputed data. For arbitrary parametric losses and data-independent imputation rules, we establish that the imputation-induced gradient bias admits a first-order expansion in the missingness vector p . We propose the Richardson-SGD procedure, which turns this structure into an algorithm by deliberately adding controlled missingness. This cancels the leading bias term, reducing gradient bias from $O(\|p\|)$ to $O(\|p\|^2)$ and the one-pass SGD error floor from $O(\|p\|^2)$ to $O(\|p\|^4)$. Our experiments show that one-step Richardson-SGD procedure successfully improves the convergence of SGD for a variety of parametric models and imputation methods. The procedure is lightweight, model-agnostic, and compatible with standard imputation pipelines. Overall, our results show that controlled additional missingness can be more than a nuisance: used carefully, it becomes a practical tool for reducing bias in stochastic learning from incomplete data.

Our debiasing procedure requires to generate more missing data with the same distribution as the original sample, but at an increased scale. Doing so is easy for independent hMCAR data, but becomes challenging in presence of anticorrelation between masks components. In this setting, we are not able to generate more missing data along all coordinates simultaneously while respecting the form of the original missing data distribution. Future research directions are to extend our procedure to such settings. Note however that, in practice, our procedure may be relatively robust to missingness misspecification (Appendix K), which leaves some hope to establish positive results in such settings.

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A Additional notation and technical preliminaries

This appendix collects notation and elementary identities used throughout the proofs.

Notations. For a mask $m \in \{0, 1\}^d$, $S(m) = \{j : m_j = 1\}$. We write $G_S(w; X, Y, \xi) := g(w; \mathcal{I}(X^{\text{NA}, S}, M_S, \xi), Y)$ for the gradient when exactly the coordinates in S are declared missing, and $G_\emptyset = g(w; X, Y)$. We use $\Delta_S G_\emptyset := \prod_{j \in S} (T_j - I) G_\emptyset$, where T_j replaces a sample by its version with coordinate j declared missing. We write $a_j(X) = 1$ under hMCAR and $a_j(X) = q_j(V)$ under sMAR.

Inclusion–exclusion identity. For every $S \subseteq [d]$, $\Delta_S G_\emptyset = \sum_{T \subseteq S} (-1)^{|S|-|T|} G_T$, and inversely $G_S = \sum_{T \subseteq S} \Delta_T G_\emptyset$ (Lemma B.1). This is the discrete-difference identity that drives the bias expansion.

Conditional independence and singleton probabilities. Under hMCAR, $\rho_{\{j\}} = p_j$. Under sMAR, $\rho_{\{j\}} = p_j q_j(V)$. Under conditional independence of the missing indicators given V , $\rho_S = \prod_{j \in S} \rho_{\{j\}}$, hence the bias is a multilinear polynomial in p (or in $p_j q_j(V)$ pointwise).

E: redéfinir ce qu'est ρ_j ici

E: Ici, je mettrais un environnement Lemme ou proposition en reprenant l'équation 7 dans le corps du texte. Si des futurs travaux veulent se référer à ce point, c'est plus facile de cette façon. La preuve peut ne faire qu'une ligne.

B Proofs for the missingness-bias expansion

This appendix proves the structural expansion of the imputation-induced gradient bias.

First, we prove a purely algebraic identity: the gradient obtained after hiding any set of coordinates can be decomposed into a sum of finite-difference effects. These effects isolate what is due to hiding one coordinate, what is due to hiding two coordinates jointly, and so on.

Second, we average this identity over the random missingness mask. This turns the finite-difference effects into a bias expansion whose coefficients are co-missingness probabilities. The first-order terms correspond to single missing coordinates; the remainder contains all simultaneous missingness effects.

Throughout this appendix, fix a parameter value $w \in \mathbb{R}^d$. We suppress the dependence on w whenever this improves readability. By definition, any imputation rule leaves a fully observed sample unchanged:

$$\mathcal{I}(X, \mathbf{0}, \xi) = X.$$

Thus, when no coordinate is declared missing, the imputed gradient equals the complete-data gradient. We assume that all finite differences introduced below are integrable. This is automatic, for instance, if the family of imputed gradients is uniformly integrable.

Gradients indexed by deterministic missingness sets. For a deterministic set $S \subseteq [d]$, let $X^{\text{NA}, S}$ be the version of X in which exactly the coordinates in S are replaced by NA. Let $M_S \in \{0, 1\}^d$ be the deterministic mask associated with S :

$$(M_S)_j = \mathbf{1}\{j \in S\}.$$

We define

$$G_S := g(w; \mathcal{I}(X^{\text{NA}, S}, M_S, \xi), Y). \quad (15)$$

Thus, G_S is the gradient we would compute if we deliberately declared exactly the coordinates in S missing and then applied the imputation rule. In particular,

$$G_\emptyset = g(w; \mathcal{I}(X, \mathbf{0}, \xi), Y) = g(w; X, Y).$$

Finite missingness differences. The objects G_S describe gradients under different missingness patterns. To separate the effect of one coordinate from the extra effect of several coordinates being missing together, we use finite differences. For every $S \subseteq [d]$, define

$$D_S := \sum_{T \subseteq S} (-1)^{|S|-|T|} G_T, \quad D_\emptyset := G_\emptyset. \quad (16)$$

The first examples are

$$D_{\{j\}} = G_{\{j\}} - G_\emptyset,$$

E: il ne faudrait pas ajouter ça dans les hypothèses des résultats, dans le corps du texte ?

513 and

$$D_{\{j,k\}} = G_{\{j,k\}} - G_{\{j\}} - G_{\{k\}} + G_{\emptyset}.$$

514 The interpretation is as follows. The term $D_{\{j\}}$ is the direct effect of hiding coordinate j . The term
 515 $D_{\{j,k\}}$ is not the full effect of hiding j and k ; it is only the additional interaction left after removing
 516 the two separate single-coordinate effects. Higher-order terms D_S have the same meaning: they
 517 isolate the part of the missingness effect that appears only when all coordinates in S are hidden
 518 together.

519 The following simple result shows that the full effect of hiding the coordinates in A can be rebuilt by
 520 adding all finite-difference effects supported inside A .

521 **Lemma B.1** (Deterministic mask expansion). *For every deterministic set $A \subseteq [d]$,*

$$G_A = \sum_{S \subseteq A} D_S. \quad (17)$$

522 *Proof.* Starting from the definition of D_S ,

$$\sum_{S \subseteq A} D_S = \sum_{S \subseteq A} \sum_{T \subseteq S} (-1)^{|S|-|T|} G_T.$$

523 We now group the terms by G_T . A fixed G_T appears only in those sums with $T \subseteq S \subseteq A$, so

$$\sum_{S \subseteq A} D_S = \sum_{T \subseteq A} G_T \sum_{S: T \subseteq S \subseteq A} (-1)^{|S|-|T|}.$$

524 For fixed $T \subseteq A$, write $S = T \cup R$, where $R \subseteq A \setminus T$. Then the inner sum becomes

$$\sum_{R \subseteq A \setminus T} (-1)^{|R|} = (1 - 1)^{|A \setminus T|},$$

525 which results from the binomial expansion of the right-hand side term. This term equals 1 if $T = A$,
 526 and 0 otherwise. Therefore, every term cancels except G_A , proving (17). $\square$

527 **From a deterministic mask to a random mask.** We now let $M \in \{0, 1\}^d$ be the actual random
 528 missingness mask. Recall that $S(M) := \{j : M_j = 1\}$ is the set of missing coordinates for the
 529 mask M . The corresponding imputed gradient is $G_{S(M)}$. For each $S \subseteq [d]$, define the conditional
 530 co-missingness probability

$$\rho_S := \mathbb{E} \left[\prod_{j \in S} M_j \mid X, Y \right] = \mathbb{P}(M_j = 1 \text{ for all } j \in S \mid X, Y), \quad \rho_{\emptyset} := 1. \quad (18)$$

531 Thus, $\rho_{\{j\}}$ is the conditional probability that coordinate j is missing, while $\rho_{\{j,k\}}$ is the conditional
 532 probability that j and k are missing simultaneously.

533 **Lemma B.2** (Random mask expansion). *Consider any data-independent imputation parametrized
 534 by ξ , as defined in (2). Let $M \in \{0, 1\}^d$ be any missingness mask. Then the corresponding imputed
 535 gradient $G_{S(M)}$ satisfies*

$$\mathbb{E} [G_{S(M)} \mid X, Y, \xi] = G_{\emptyset} + \sum_{\emptyset \neq S \subseteq [d]} \rho_S D_S. \quad (19)$$

536 *Proof.* Apply Lemma B.1 to the random set $A = S(M)$. For a fixed realization of the mask,

$$G_{S(M)} = \sum_{S \subseteq S(M)} D_S.$$

537 The condition $S \subseteq S(M)$ is equivalent to saying that every coordinate in S is missing, namely
 538 $M_j = 1$ for all $j \in S$. Therefore

$$\mathbf{1}\{S \subseteq S(M)\} = \prod_{j \in S} M_j,$$

539 which yields

$$G_{S(M)} = \sum_{S \subseteq [d]} \left(\prod_{j \in S} M_j \right) D_S.$$

540 Conditional on (X, Y, ξ) , the finite differences D_S are fixed, and only the mask remains random.
 541 Moreover, by assumption, we have $\xi \perp M \mid (X, Y)$, which leads to

$$\mathbb{E} \left[\prod_{j \in S} M_j \mid X, Y, \xi \right] = \mathbb{E} \left[\prod_{j \in S} M_j \mid X, Y \right] = \rho_S.$$

542 Hence

$$\mathbb{E} [G_{S(M)} \mid X, Y, \xi] = \sum_{S \subseteq [d]} \rho_S D_S.$$

543 The term $S = \emptyset$ equals $D_\emptyset = G_\emptyset$, which gives (19). $\square$

544 **Bias expansion.** We can now prove the first-order structure of the gradient bias. The random
 545 imputed gradient used by the learner is

$$\hat{g}(w) = G_{S(M)}.$$

546 Since $G_\emptyset = g(w; X, Y)$, we have $\mathbb{E}[G_\emptyset] = \nabla L(w)$. Taking expectations in Lemma B.2 therefore
 547 yields

$$\mathcal{B}(w, p) := \mathbb{E}[\hat{g}(w)] - \nabla L(w) = \sum_{\emptyset \neq S \subseteq [d]} \mathbb{E}[\rho_S D_S]. \quad (20)$$

548 This identity is the key building block. It says that the bias is a sum over all nonempty missingness
 549 sets S . Each term has two factors:

- 550 • ρ_S , the probability that all coordinates in S are missing;
- 551 • D_S , the incremental gradient effect created by hiding exactly the coordinates in S , after
 552 lower-order effects have been subtracted.

553 Thus, singletons $S = \{j\}$ produce the first-order bias, while sets with $|S| \geq 2$ produce the co-
 554 missingness remainders.

555 *Proof of Proposition 1.* Start from the exact expansion (20). We separate the singleton terms from
 556 the terms involving at least two missing coordinates:

$$\mathcal{B}(w, p) = \sum_{j=1}^d \mathbb{E}[\rho_{\{j\}} D_{\{j\}}] + \sum_{|S| \geq 2} \mathbb{E}[\rho_S D_S]. \quad (21)$$

557 We now identify the singleton probabilities under the mechanisms considered in the paper.

558 Under hMCAR,

$$\rho_{\{j\}} = \mathbb{P}(M_j = 1) = p_j.$$

559 Under sMAR, with $V = X^{(\mathcal{O})}$,

$$\rho_{\{j\}} = \mathbb{P}(M_j = 1 \mid V) = p_j q_j(V).$$

560 Both cases can be written as

$$\rho_{\{j\}} = p_j a_j(X), \quad \text{with } a_j(X) = \begin{cases} 1, & \text{hMCAR,} \\ q_j(V), & \text{sMAR.} \end{cases} \quad (22)$$

561 Substituting (22) into the singleton part of (21) gives

$$\sum_{j=1}^d \mathbb{E}[\rho_{\{j\}} D_{\{j\}}] = \sum_{j=1}^d p_j \mathbb{E}[a_j(X) D_{\{j\}}].$$

562 Since $D_{\{j\}} = G_{\{j\}} - G_{\emptyset}$, this is exactly $\mathcal{A}(w)p$, where the j -th column of $\mathcal{A}(w)$ is

$$\mathcal{A}_j(w) = \mathbb{E} [a_j(X) \{G_{\{j\}}(w; X, Y, \xi) - G_{\emptyset}(w; X, Y, \xi)\}].$$

563 The remaining terms are precisely the co-missingness remainder:

$$\mathcal{R}(w, p) := \sum_{|S| \geq 2} \mathbb{E}[\rho_S D_S]. \quad (23)$$

564 Combining the singleton part and the remainder proves

$$\mathcal{B}(w, p) = \mathcal{A}(w)p + \mathcal{R}(w, p).$$

565

□

566 **Remark 1** (What the remainder contains). *The remainder $\mathcal{R}(w, p)$ is the sum of all interaction terms*
 567 *caused by simultaneous missingness. For example, the pair $\{j, k\}$ contributes*

$$\mathbb{E}[\rho_{\{j,k\}} D_{\{j,k\}}].$$

568 *If M_j and M_k are independent and each is missing with probability of order p , then $\rho_{\{j,k\}}$ is of order*
 569 *p^2 . If instead the two coordinates are always missing together, then $\rho_{\{j,k\}}$ can be of order p . Thus,*
 570 *without a weak-dependence condition on co-missingness probabilities, $\mathcal{R}(w, p)$ may be a first-order*
 571 *term, proportional to p .*

572 *Proof of Corollary 6.* Assume that the missingness indicators are conditionally independent given
 573 the variables driving the missingness mechanism. In hMCAR, this is ordinary independence. In
 574 sMAR, this is conditional independence given V .

575 Then, for every $S \subseteq [d]$, the probability that all coordinates in S are missing factorizes:

$$\rho_S = \prod_{j \in S} \rho_{\{j\}} = \prod_{j \in S} p_j a_j(X).$$

576 Substituting this into the exact remainder (23),

$$\mathcal{R}(w, p) = \sum_{|S| \geq 2} \left(\prod_{j \in S} p_j \right) \mathbb{E} \left[\left(\prod_{j \in S} a_j(X) \right) D_S \right].$$

577 Every term in this sum contains at least two factors p_j . Since the dimension is fixed and the finite
 578 differences are integrable, there exists a finite constant C_w , depending on w but not on p , such that

$$\|\mathcal{R}(w, p)\| \leq C_w \sum_{|S| \geq 2} \prod_{j \in S} p_j.$$

579 The last sum is $O(\|p\|^2)$ as $\|p\| \rightarrow 0$, because each product contains at least two entries of p .
 580 Therefore

$$\|\mathcal{R}(w, p)\| = O(\|p\|^2), \quad \|\mathcal{B}(w, p) - \mathcal{A}(w)p\| = O(\|p\|^2).$$

581

□

582 **Multilinear form under independent masks.** Under the same conditional-independence as-
 583 sumption, the whole bias expansion can be written as a multilinear polynomial in the missingness
 584 probabilities:

$$\mathcal{B}(w, p) = \sum_{\emptyset \neq S \subseteq [d]} \left(\prod_{j \in S} p_j \right) \mu_S(w), \quad (24)$$

585 where

$$\mu_S(w) := \mathbb{E} \left[\left(\prod_{j \in S} a_j(X) \right) D_S(w) \right]. \quad (25)$$

586 The coefficient $\mu_S(w)$ is the average $|S|$ -way missingness interaction: it is the effect of declaring
 587 all coordinates in S missing, after all lower order effects have been removed by inclusion-exclusion.
 588 This formula is the reason Richardson extrapolation applies: the bias is organized by powers of the
 589 missingness scale.

E: peut-être introduire la notation μ_S dans la preuve ci-dessus, pour supprimer les équations (24) et (25) et faire référence à la preuve du Crolaire 6 dans le texte pour justifier (7)

F: Yes, je vais faire la modif.

C Proofs for Richardson correction

This appendix collects the proofs of the Richardson-extrapolation results: the joint law of the further-thinned mask, first- and higher-order bias cancellation, the subset-based variant, the plug-in mechanism, and the linear-regression case study. We close with explicit GLM bias formulas.

C.1 Joint law of the further-thinned mask

Fix $C > 1$ and assume $C p_j a_j(X) \leq 1$ a.s. for every $j \in \mathcal{O}^c$, with a_j as in (22). For $j \in \mathcal{O}^c$, draw r_j as in (9), conditionally independent across j given $(X, M^{(p)})$. Define $\omega_j^{(p)} := 1 - M_j^{(p)}$ (the indicator of observation under $M^{(p)}$) and $\omega_j^{(Cp)} := \omega_j^{(p)} r_j$.

E: On peut tout réécrire avec M_j plutôt que ω_j ?

Since the imputation rule is conditionally independent of $M^{(p)}$ given (X, Y) , by construction we have $\mathbb{P}(\omega_j^{(p)} = 1 \mid X, Y) = 1 - p_j a_j(X)$, where $V := X^{(\mathcal{O})}$. Conditioning further on $M^{(p)} = m$ in the event $\omega_j^{(p)} = 1$,

F: Oui, je vais modifier

$$\begin{aligned} \mathbb{P}(\omega_j^{(Cp)} = 1 \mid X, Y) &= \mathbb{P}(\omega_j^{(p)} = 1 \mid X, Y) \mathbb{P}(r_j = 1 \mid X, Y, \omega_j^{(p)} = 1) \\ &= (1 - p_j a_j(X)) \frac{1 - C p_j a_j(X)}{1 - p_j a_j(X)} \\ &= 1 - C p_j a_j(X). \end{aligned}$$

Due to the conditional independence of $M^{(p)}$ given V , we obtain the conditional independence of $\{M_j^{(Cp)}\}_{j \in \mathcal{O}^c}$ given V . Hence, $M^{(Cp)}$ has the same conditional law as an independent mask drawn at scale Cp .

E: Faire intervenir V ici plutôt que X quand c'est possible

C.2 Proof of Proposition 2

F: Je suis d'accord, je vais modifier.

Apply Proposition 1 at scales p and Cp . Both biases admit the decomposition $\mathcal{B}(w, \cdot) = \mathcal{A}(w) \cdot + \mathcal{R}(w, \cdot)$, with the same operator $\mathcal{A}(w)$ (since by (5), $\mathcal{A}$ does not depend on p). Substituting into (10),

$$\begin{aligned} \mathbb{E}[\hat{g}_C^R(w)] - \nabla L(w) &= \frac{C \mathcal{B}(w, p) - \mathcal{B}(w, Cp)}{C - 1} \\ &= \frac{C \mathcal{A}(w) p - \mathcal{A}(w) (Cp)}{C - 1} + \frac{C \mathcal{R}(w, p) - \mathcal{R}(w, Cp)}{C - 1}. \end{aligned}$$

Linearity of $\mathcal{A}$ yields $C \mathcal{A}(w) p - \mathcal{A}(w) (Cp) = 0$, so only the remainder survives:

$$\mathbb{E}[\hat{g}_C^R(w)] - \nabla L(w) = \frac{C \mathcal{R}(w, p) - \mathcal{R}(w, Cp)}{C - 1}.$$

Under conditional independence of the $\{M_j\}_{j \in \mathcal{O}^c}$ given V , Proposition 1 gives $\|\mathcal{R}(w, p)\| = O(\|p\|^2)$ and, by the same bound applied at scale Cp , $\|\mathcal{R}(w, Cp)\| = O(C^2 \|p\|^2) = O(\|p\|^2)$. Combining,

$$\|\mathbb{E}[\hat{g}_C^R(w)] - \nabla L(w)\| = O(\|p\|^2),$$

which is (11). Without conditional independence, the same calculation gives only $\|\mathcal{R}(w, p)\|, \|\mathcal{R}(w, Cp)\| = o(\|p\|)$, hence the residual bias is also $o(\|p\|)$, but the leading $\mathcal{A}(w) p$ term is still cancelled exactly. $\square$

E: Sans l'indépendance, le reste n'est plus forcément $o(p)$ donc le résultat ne tient plus tel quel.

C.3 Proof of Corollary 2 (higher-order cancellation)

Under independent masks, $\mathcal{B}(w, p) = \sum_{\emptyset \neq S \subseteq [d]} \left(\prod_{j \in S} p_j \right) \mu_S(w)$ from (7). Group terms by $|S|$:

$$\mathcal{B}(w, p) = \sum_{m=1}^d \beta_m(w, p), \quad \text{with} \quad \beta_m(w, p) := \sum_{|S|=m} \left(\prod_{j \in S} p_j \right) \mu_S(w),$$

so that $\beta_m(w, \cdot)$ is homogeneous of degree m , i.e. $\beta_m(w, Cp) = C^m \beta_m(w, p)$. For a sequence of expansion factors $1 = C_0 < C_1 < \dots < C_k$ with $C_k p_j a_j(X) \leq 1$ a.s.,

$$\mathcal{B}(w, C_\ell p) = \sum_{m=1}^d C_\ell^m \beta_m(w, p), \quad \ell = 0, \dots, k.$$

F: Pardon oui c'est faux comme c'est écrit, je vais écrire quelque chose de propre juste pour dire: si c'est pas indépendant, le reste est au pire de la forme linéaire plus $o(p)$ et donc si linéaire, ce terme par dans richardson, et sinon on a quand même $o(p)$ à la fin.

618 The Vandermonde system

$$\sum_{\ell=0}^k \alpha_{\ell} = 1, \quad \sum_{\ell=0}^k \alpha_{\ell} C_{\ell}^m = 0, \quad m = 1, \dots, k,$$

619 admits a unique solution $\alpha \in \mathbb{R}^{k+1}$ since the matrix $(C_{\ell}^m)_{\ell,m=0}^k$ is a non-singular Vandermonde.
 620 With this choice of α ,

$$\begin{aligned} \mathbb{E} [\hat{g}^{[k]}(w)] - \nabla L(w) &= \sum_{\ell=0}^k \alpha_{\ell} \mathcal{B}(w, C_{\ell} p) \\ &= \sum_{m=1}^d \left(\sum_{\ell=0}^k \alpha_{\ell} C_{\ell}^m \right) \beta_m(w, p) \\ &= \sum_{m=k+1}^d \left(\sum_{\ell=0}^k \alpha_{\ell} C_{\ell}^m \right) \beta_m(w, p), \end{aligned}$$

621 where the last equality uses $\sum_{\ell} \alpha_{\ell} C_{\ell}^m = 0$ for $m = 1, \dots, k$ and $\sum_{\ell} \alpha_{\ell} C_{\ell}^0 = 1$ but the
 622 $m = 0$ term contributes $\sum_{\ell} \alpha_{\ell} \mathcal{B}(w, 0) = 0$ since $\mathcal{B}(w, 0) = 0$. Each $\beta_m(w, p)$ is bounded by
 623 $\|p\|_{\infty}^m \sum_{|S|=m} \|\mu_S(w)\| = O(\|p\|^m)$, hence

$$\|\mathbb{E}[\hat{g}^{[k]}(w)] - \nabla L(w)\| = O(\|p\|^{k+1}).$$

624 Finally, let $d_{\text{miss}} = \#\{j : p_j > 0\}$. When $k = d_{\text{miss}}$, every S with $|S| > d_{\text{miss}}$ has at least one
 625 coordinate with $p_j = 0$, so $\prod_{j \in S} p_j = 0$ and $\beta_m(w, p) = 0$ for $m > d_{\text{miss}}$. The residual bias
 626 vanishes identically: $\mathbb{E}[\hat{g}^{[d_{\text{miss}}]}(w)] = \nabla L(w)$. The argument under sMAR is identical, with p_j
 627 replaced by $p_j a_j(X)$ inside the expectation defining $\mu_S(w)$. $\square$

628 C.4 Proof of Proposition 3 (plug-in mechanism)

629 Let $\lambda_j(V) := p_j q_j(V)$ and $\hat{\lambda}_j(V) := \hat{p}_j \hat{q}_j(V)$. We write M^R for the further-thinned mask produced
 630 by the plug-in rule with intensities $\hat{\lambda}_j$.

631 **Step 1: Effective intensity of the plug-in further-thinned mask.** Conditioning on V and using
 632 the conditional independence of r_j and $M^{(p)}$ given V ,

$$\begin{aligned} \mathbb{P}(M_j^R = 1 \mid V) &= 1 - \mathbb{P}(\omega_j^{(p)} = 1, r_j = 1 \mid V) \\ &= 1 - (1 - \lambda_j(V)) \frac{1 - C \hat{\lambda}_j(V)}{1 - \hat{\lambda}_j(V)}. \end{aligned}$$

633 Define the *effective intensity* after plug-in thinning by

$$\tilde{\lambda}_j(V) := \mathbb{P}(M_j^R = 1 \mid V).$$

634 Expanding the previous display over the common denominator $1 - \hat{\lambda}_j(V)$, we obtain

$$\begin{aligned} \tilde{\lambda}_j(V) &= \frac{1 - \hat{\lambda}_j(V) - (1 - \lambda_j(V))(1 - C \hat{\lambda}_j(V))}{1 - \hat{\lambda}_j(V)} \\ &= \frac{\lambda_j(V) + (C - 1) \hat{\lambda}_j(V) - C \lambda_j(V) \hat{\lambda}_j(V)}{1 - \hat{\lambda}_j(V)} \\ &= \frac{C \lambda_j(V) (1 - \hat{\lambda}_j(V)) + (C - 1) (\hat{\lambda}_j(V) - \lambda_j(V))}{1 - \hat{\lambda}_j(V)} \\ &= C \lambda_j(V) + (C - 1) \frac{\hat{\lambda}_j(V) - \lambda_j(V)}{1 - \hat{\lambda}_j(V)}. \end{aligned} \tag{26}$$

E: je ne comprends pas pourquoi il y a des *hat* ici, on devrait retrouver λ car on a écrit une proba et non quelque chose qui est un estimateur

F: On rajoute artificiellement des Nans, avec une bernouilli, donc la proba depend de notre estimation de λ_j , si on avait la valeur exacte, alors les *hat* seraient les λ_j . Exemple typique: on estimation $p_1 \simeq 0.20$, on veut doubler la proba, on simule $B(1 - 0.40/0.20)$, la proba de manquante est donc $p_1 \cdot 0.2 \simeq$

E: ah oui, d'accord! Peut-être introduire $\hat{r}_j$ qui depend de $\hat{\lambda}_j$, en rappelant son expression pour que la différence entre les quantités estimées et théorique soient claires

635 The first term is the desired intensity of a new draw at scale Cp ; the second is the plug-in error.
 636 Setting

$$e_j^{(C)}(V) := \tilde{\lambda}_j(V) - C \lambda_j(V) = (C-1) \frac{\hat{\lambda}_j(V) - \lambda_j(V)}{1 - \hat{\lambda}_j(V)},$$

637 we have $\tilde{\lambda}(V) = C\lambda(V) + e^{(C)}(V)$.

638 **Step 2: Bias of the plug-in Richardson gradient.** By Proposition 1 applied at scale λ for the
 639 unthinned gradient and at the (random) effective scale $\tilde{\lambda}$ for the further-thinned gradient,

$$\begin{aligned} \mathbb{E}[\hat{g}_{C,\tilde{\lambda}}^R(w)] - \nabla L(w) &= \frac{C \mathcal{B}(w, p) - \mathbb{E}[\mathcal{B}(w, \tilde{p})]}{C-1} \\ &= \frac{C \mathcal{A}(w) p - \mathbb{E}[\mathcal{A}(w) (Cp + e^{(C)}(V)/\bar{a})]}{C-1} \\ &\quad + \frac{C \mathcal{R}(w, p) - \mathbb{E}[\mathcal{R}(w, Cp + e^{(C)}(V)/\bar{a})]}{C-1}, \end{aligned}$$

640 where, with a slight abuse of notation, we have absorbed the conditional intensity $a_j(X)$ into the
 641 marginal scale (the multiplicative factor $\bar{a}$ cancels through the linearity of $\mathcal{A}$). By linearity of the
 642 operator $\mathcal{A}$,

$$\frac{C \mathcal{A}(w) p - \mathcal{A}(w) (Cp + e^{(C)})}{C-1} = -\frac{1}{C-1} \mathcal{A}(w) e^{(C)},$$

643 and the remainder is $O(\|p\|^2 + \|p\| \|e^{(C)}\| + \|e^{(C)}\|^2)$. Combining,

$$\|\mathbb{E}[\hat{g}_{C,\tilde{\lambda}}^R(w)] - \nabla L(w)\| \leq \frac{1}{C-1} \|\mathcal{A}(w)\| \|e^{(C)}\| + O(\|p\|^2 + \|p\| \|e^{(C)}\| + \|e^{(C)}\|^2). \quad (27)$$

644 The operator norm $\|\mathcal{A}(w)\|$ is bounded by $G_\star$ (the assumed L^2 bound on imputation gradient gaps):
 645 indeed, for any j ,

$$\|\mathcal{A}_j(w)\| = \|\mathbb{E}[a_j(X)(G_{\{j\}} - G_\emptyset)]\| \leq \mathbb{E}[a_j(X) \|G_{\{j\}} - G_\emptyset\|] \leq G_\star,$$

646 since $0 < a_j \leq 1$.

647 **Step 3: Bound on the plug-in error.** By the elementary inequality $|\hat{p}_j \hat{q}_j(V) - p_j q_j(V)| \leq$
 648 $|\hat{p}_j - p_j| |q_j(V)| + |\hat{p}_j| |\hat{q}_j(V) - q_j(V)|$ and the bounds $|q_j(V)| \leq 1$, $|\hat{p}_j| \leq \|p\|_\infty + \delta_p$,

$$|\hat{\lambda}_j(V) - \lambda_j(V)| \leq \delta_p + (\|p\|_\infty + \delta_p) \delta_q.$$

649 Since $\hat{\lambda}_j(V) \leq \rho < 1$,

$$|e_j^{(C)}(V)| \leq \frac{C-1}{1-\rho} [\delta_p + (\|p\|_\infty + \delta_p) \delta_q],$$

650 hence

$$\|e^{(C)}\| = O(\delta_p + \|p\|_\infty \delta_q + \delta_p \delta_q).$$

651 **Step 4: Concluding.** Substituting the bound on $\|e^{(C)}\|$ into (27) and simplifying,

$$\|\mathbb{E}[\hat{g}_{C,\tilde{\lambda}}^R(w)] - \nabla L(w)\| = O(\|p\|^2 + \delta_p + \|p\|_\infty \delta_q + \delta_p \delta_q),$$

652 where the implicit constants depend only on C , ρ , and $G_\star$. In the MCAR case $q_j \equiv 1$, $\delta_q = 0$ and
 653 the bound collapses to $O(\|p\|^2 + \delta_p)$. $\square$

654 D Bias formulas for specific generalized linear models

655 We record explicit expressions for the leading-order population bias $\mathcal{A}(w) p$ in three GLMs under
 656 heterogeneous MCAR with zero imputation. Notably, linear regression has at most order 2 bias,
 657 while logistic and Poisson regression have, in general, full-order bias (up to order d).

E: pas génial car il faut introduire $\bar{a}$. On ne peut pas s'en passer et écrire la dépendance en a_j explicitement?

F: Si effectivement, je vais réécrire ça mieux

E: independent? Sinon il y a possiblement un terme linéaire en p dans le reste ?

F: Oui on peut écrire indépendant. Je voulais juste dire que c'est le terme $\mathcal{A}$ dans tous les cas, indépen-

658 **Linear regression (squared loss).** For $\ell(w; x, y) = \frac{1}{2}(w^\top x - y)^2$ and zero imputation,

$$(\mathcal{A}(w)p)_j = -p_j \nabla_j L(w) - \sum_{k \neq j} p_k S_{jk} w_k, \quad S := \mathbb{E}[X X^\top].$$

659 The detailed derivation, including the exact non-asymptotic version, is reproduced in Appendix E.

660 **Logistic regression.** For $\ell(w; x, y) = \log(1 + e^{-y w^\top x})$ with $y \in \{-1, +1\}$, the gradient is
 661 $g(w; x, y) = -y \sigma(-y w^\top x) x$ where σ is the logistic function. Under zero imputation and heteroge-
 662 neous MCAR,

$$(\mathcal{A}(w)p)_j = -p_j \nabla_j L(w) - \sum_{k \neq j} p_k \mathbb{E} \left[Y (\sigma(-Y w^\top X) - \sigma(-Y w^\top X^{(-k)})) X_j \right],$$

663 where $X^{(-k)}$ is X with X_k replaced by 0.

664 **Poisson regression.** For $\ell(w; x, y) = e^{w^\top x} - y w^\top x$, the gradient is $g(w; x, y) = (e^{w^\top x} - y) x$,
 665 hence

$$(\mathcal{A}(w)p)_j = -p_j \nabla_j L(w) - \sum_{k \neq j} p_k \mathbb{E} \left[(e^{w^\top X^{(-k)}} - e^{w^\top X}) X_j \right].$$

666 All three expressions are obtained by substituting the corresponding loss into (5). They share the
 667 same structural form: a coordinate-wise diagonal contribution $-p_j \nabla_j L(w)$ that vanishes at w^* , plus
 668 an off-diagonal correction that, in general, does not.

669 E Linear regression: a transparent case study

670 The goal of this appendix is to show on the simplest GLM that, under heterogeneous MCAR with
 671 independent masks and zero imputation, the population gradient bias is a polynomial of degree at
 672 most 2 in p . By Corollary 2, second-order Richardson with two factors $C_1 < C_2$ therefore cancels
 673 this bias *exactly*, while a single Richardson step already reduces it from $O(\|p\|)$ to $O(\|p\|^2)$.

674 **Setting.** We work at the single-observation level with squared loss,

$$\ell(w; x, y) = \frac{1}{2}(w^\top x - y)^2, \quad g(w; x, y) = (w^\top x - y) x.$$

675 The population risk is $L(w) = \frac{1}{2} \mathbb{E}[(w^\top X - Y)^2]$ with $\nabla L(w) = S w - b$, $S := \mathbb{E}[X X^\top]$, $b :=$
 676 $\mathbb{E}[Y X]$. We assume heterogeneous MCAR with independent mask coordinates, $\mathbb{P}(M_j = 1) = p_j$,
 677 and zero imputation $\tilde{X} = (1 - M) \odot X$. The imputed gradient is $\hat{g}(w) := g(w; \tilde{X}, Y) = (w^\top \tilde{X} -$
 678 $Y) \tilde{X}$.

679 E.1 Sample-conditional and population biases

680 **Proposition 4** (Sample-conditional bias). *For each $j \in [d]$,*

$$\mathbb{E}_M \left[\hat{g}_j(w; \tilde{X}, Y) \mid X, Y \right] - g_j(w; X, Y) = -p_j X_j^2 w_j - \sum_{k \neq j} (p_j + p_k - p_j p_k) X_j X_k w_k + p_j Y X_j, \quad (28)$$

682 or equivalently

$$\mathbb{E}_M \left[\hat{g}_j(w; \tilde{X}, Y) \mid X, Y \right] - g_j(w; X, Y) = -p_j g_j(w; X, Y) - (1 - p_j) \sum_{k \neq j} p_k X_j X_k w_k. \quad (29)$$

683 *Proof.* With $\omega_j := 1 - M_j$ and $\tilde{X}_j = \omega_j X_j$, the imputed j -th gradient is

$$\hat{g}_j(w; \tilde{X}, Y) = (w^\top \tilde{X} - Y) \tilde{X}_j = \left(\sum_k w_k \omega_k X_k - Y \right) \omega_j X_j.$$

E: pourquoi est-ce important, le fait que le terme s'annule en w^* ? On ne débiaise pas le gradient à l'optimum. Si c'est important, il faudrait faire cette remarque pour la régression linéaire ci-dessus aussi

F: Ça ne l'est pas forcément, c'était plutôt une remarque.

E: Remettre les hypothèses dans chacune des propositions ci-dessous "independent heterogeneous MCAR".

684 Conditioning on (X, Y) and using independence of the mask coordinates ($\mathbb{E}[\omega_j] = \mathbb{E}[\omega_j^2] = 1 - p_j$
 685 and $\mathbb{E}[\omega_j \omega_k] = (1 - p_j)(1 - p_k)$ for $k \neq j$),

$$\mathbb{E}_M [\hat{g}_j(w; \tilde{X}, Y) \mid X, Y] = (1 - p_j) X_j^2 w_j + \sum_{k \neq j} (1 - p_j)(1 - p_k) X_j X_k w_k - (1 - p_j) Y X_j.$$

686 Subtracting the complete-data gradient $g_j(w; X, Y) = X_j^2 w_j + \sum_{k \neq j} X_j X_k w_k - Y X_j$ yields (28)
 687 after expanding $(1 - p_j)(1 - p_k) - 1 = -(p_j + p_k - p_j p_k)$. To obtain (29), factor $-p_j$ in front of
 688 $g_j(w; X, Y)$:

$$\begin{aligned} & -p_j X_j^2 w_j - p_j \sum_{k \neq j} X_j X_k w_k + p_j Y X_j - (1 - p_j) \sum_{k \neq j} p_k X_j X_k w_k \\ & = -p_j g_j(w; X, Y) - (1 - p_j) \sum_{k \neq j} p_k X_j X_k w_k. \end{aligned} \quad \square$$

689 **Corollary 4** (Population bias of zero-imputed linear regression). *With $B_j(w; p) :=$*
 690 $\mathbb{E} [\hat{g}_j(w; \tilde{X}, Y)] - \nabla_j L(w)$,

$$B_j(w; p) = -p_j \nabla_j L(w) - (1 - p_j) \sum_{k \neq j} p_k S_{jk} w_k, \quad (30)$$

691 hence $\|B(w; p)\| = O(\|p\|)$.

692 *Proof.* Take expectation in (29) and use $\mathbb{E}[g_j(w; X, Y)] = \nabla_j L(w)$ and $\mathbb{E}[X_j X_k] = S_{jk}$ to
 693 obtain (30). The norm bound follows from $|B_j(w; p)| \leq p_j |\nabla_j L(w)| + \sum_{k \neq j} p_k |S_{jk} w_k| \leq$
 694 $\|p\|_\infty (|\nabla_j L(w)| + \sum_k |S_{jk} w_k|)$. $\square$

695 E.2 Polynomial structure and exact debiasing in two Richardson steps

696 We now make explicit that the population gradient bias is a polynomial of degree at most 2 in p ,
 697 hence is annihilated exactly by second-order Richardson with two factors.

698 **The bias is degree-2 in p .** From (30),

$$B_j(w; p) = \underbrace{\left(-p_j \nabla_j L(w) - \sum_{k \neq j} p_k S_{jk} w_k \right)}_{=: L_j(w; p), \text{ linear in } p} + \underbrace{p_j \sum_{k \neq j} p_k S_{jk} w_k}_{=: Q_j(w; p), \text{ quadratic in } p}. \quad (31)$$

699 The same conclusion follows from the general expansion (7) of Section 3, since for the squared loss
 700 the iterated finite differences $\Delta_S G_\varnothing$ vanish identically for $|S| \geq 3$ (the gradient is bilinear in X).

701 One Richardson step removes the linear part.

702 **Proposition 5** (First-order Richardson cancellation, heterogeneous squared-loss MCAR). *For $C > 1$*
 703 *with $C p_j < 1$ for every j , the first-order Richardson gradient (10) satisfies*

$$\mathbb{E} [\hat{g}_j^R(w)] - \nabla_j L(w) = -C p_j \sum_{k \neq j} p_k S_{jk} w_k = -C Q_j(w; p).$$

704 *In particular, $\|\mathbb{E}[\hat{g}^R(w)] - \nabla L(w)\| = O(\|p\|^2)$, while the uncorrected bias is $O(\|p\|)$. If $p_j = 0$*
 705 *for some j , the Richardson bias vanishes on coordinate j .*

706 *Proof.* Apply $\mathbb{E}[\hat{g}_j^R] - \nabla_j L = (C B_j(w; p) - B_j(w; Cp))/(C - 1)$ to (31). Using

$$L_j(w; Cp) = C L_j(w; p) \quad \text{and} \quad Q_j(w; Cp) = C^2 Q_j(w; p),$$

707 we have

$$\begin{aligned} \frac{C B_j(w; p) - B_j(w; Cp)}{C - 1} &= \frac{C L_j(w; p) - C L_j(w; p)}{C - 1} + \frac{C Q_j(w; p) - C^2 Q_j(w; p)}{C - 1} \\ &= -C Q_j(w; p), \end{aligned}$$

708 which gives the claim. $\square$

709 **Two Richardson steps cancel the bias exactly.**

710 **Corollary 5** (Exact Richardson debiasing for linear regression). *Let $1 = C_0 < C_1 < C_2$ be three*
 711 *expansion factors with $C_2 p_j < 1$ for every j . The unique Vandermonde solution*

$$(\alpha_0, \alpha_1, \alpha_2) \text{ of } \begin{cases} \alpha_0 + \alpha_1 + \alpha_2 = 1, \\ \alpha_0 + \alpha_1 C_1 + \alpha_2 C_2 = 0, \\ \alpha_0 + \alpha_1 C_1^2 + \alpha_2 C_2^2 = 0, \end{cases}$$

712 *gives a second-order Richardson gradient $\hat{g}^{[2]} = \sum_{\ell} \alpha_{\ell} \hat{g}^{(C_{\ell} p)}$ with $\mathbb{E}[\hat{g}^{[2]}(w)] = \nabla L(w)$ exactly.*
 713 *The same statement holds in independent sMAR (conditional on V), after replacing p_j by $p_j a_j(X)$*
 714 *inside the expectations defining L_j and Q_j .*

715 *Proof.* The bias (31) is a polynomial of degree ≤ 2 in p with no constant term, so $\mathcal{B}(w, Cp) =$
 716 $C L(w; p) + C^2 Q(w; p)$ for any $C > 0$. Applying $\sum_{\ell} \alpha_{\ell} \mathcal{B}(w, C_{\ell} p)$ and using the Vandermonde
 717 conditions, both L and Q contributions vanish. $\square$
 718

719 This recovers, in our framework, the closed-form debiasing of Sportisse et al. [32] for linear regression
 720 with squared loss under independent MCAR, and extends it to the sMAR mechanisms of Section 2,
 721 where no closed-form bias is available.

722 E.3 Homogeneous MCAR illustration

724 When $p_1 = \dots = p_d = p$, the bias (7) reduces to a polynomial in the scalar p :

$$\mathcal{B}(w, p) = p A(w) + p^2 Q(w) + \dots + p^d D(w),$$

725 where $A(w) = \sum_j \mu_j(w)$, $Q(w) = \sum_{j < k} \mu_{jk}(w)$, etc. The Richardson estimator with $C = 2$ thus
 726 satisfies

$$\mathbb{E}[\hat{g}_2^R(w)] - \nabla L(w) = 2(pA + p^2Q) - (2pA + 4p^2Q) + O(p^3) = -2p^2Q(w) + O(p^3).$$

727 For squared loss this matches the expression derived in Appendix E.

E: Cela ressemble plutôt à un système $V(1, C_1, C_2)^T \alpha = (1, 0, 0)$ donc à une transposée de Vandermonde ? ça ne change pas le fait que le système admette une unique solution car $\det(V) = \det(V^T)$. Peut-être le mentionner ?

E: bon, en fait, je ne sais pas où le mettreg. Tu es sûr que tu veux garder ce résultat ?

728 F One-pass biased-SGD convergence consequences

729 We provide a simple proof for the non-averaged iterates of SGD with biased gradients. The rates
 730 stated in the main text are then recovered by applying this result with the bias corresponding to each
 731 method. Note that one could also aim for similar guarantees for averaged SGD with a broader class
 732 of step sizes, namely $\eta_t \propto t^{-a}$ with $a \in (1/2, 1)$, which notably does not require prior knowledge of
 733 the strong convexity constant of the loss [27, 4]. We do not pursue this direction here, as our main
 734 focus is bias reduction, and the theorem below already illustrates its practical benefit.

735 **Proposition 6** (One-pass SGD with Bias). *Assume L is α -strongly convex and β -smooth, and let w^**
 736 *be its unique minimizer. Run one-pass SGD over n i.i.d. samples,*

$$w_{i+1} = w_i - \eta_i \hat{g}_i(w_i),$$

737 *with step sizes*

$$\eta_i = \frac{c}{i + \gamma}, \quad \alpha c > 1,$$

738 *where γ is large enough that*

$$\eta_i \leq \frac{\alpha}{6\beta^2} \quad \text{for all } i.$$

739 *Suppose that the imputed gradient satisfies, uniformly along the trajectory,*

$$\|\mathbb{E}[\hat{g}_i(w_i) \mid w_i] - \nabla L(w_i)\| \leq B(p),$$

740 *and*

$$\mathbb{E}[\|\hat{g}_i(w_i) - \mathbb{E}[\hat{g}_i(w_i) \mid w_i]\|^2 \mid w_i] \leq \sigma^2 \quad \text{a.s.}$$

741 *Then*

$$\mathbb{E}\|w_n - w^*\|^2 = O(B(p)^2) + O\left(\frac{\sigma^2}{n}\right).$$

742 *In particular, if $\sigma^2 = O(1)$, then*

$$\mathbb{E}\|w_n - w^*\|^2 = O(B(p)^2) + O(1/n). \quad (32)$$

E: there exists $B(p)$ such that

743 *Proof.* Write

$$\mathbb{E}[\hat{g}_i(w_i) \mid w_i] = \nabla L(w_i) + \Delta_i, \quad \|\Delta_i\| \leq B(p).$$

744 Also write

$$\hat{g}_i(w_i) = \nabla L(w_i) + \Delta_i + \xi_i, \quad \mathbb{E}[\xi_i \mid w_i] = 0, \quad \mathbb{E}[\|\xi_i\|^2 \mid w_i] \leq \sigma^2.$$

745 Let

$$\delta_i := \mathbb{E}\|w_i - w^*\|^2.$$

746 Since w^* minimizes L , $\nabla L(w^*) = 0$. Expanding one SGD step gives

$$\begin{aligned} \mathbb{E}[\|w_{i+1} - w^*\|^2 \mid w_i] &= \|w_i - w^*\|^2 - 2\eta_i \langle w_i - w^*, \nabla L(w_i) \rangle \\ &\quad - 2\eta_i \langle w_i - w^*, \Delta_i \rangle + \eta_i^2 \mathbb{E}[\|\nabla L(w_i) + \Delta_i + \xi_i\|^2 \mid w_i]. \end{aligned}$$

747 By strong convexity,

$$\langle w_i - w^*, \nabla L(w_i) \rangle \geq \alpha \|w_i - w^*\|^2.$$

748 By Young's inequality,

$$2|\langle w_i - w^*, \Delta_i \rangle| \leq \frac{\alpha}{2} \|w_i - w^*\|^2 + \frac{2}{\alpha} B(p)^2.$$

749 By smoothness and $\nabla L(w^*) = 0$,

$$\|\nabla L(w_i)\| \leq \beta \|w_i - w^*\|.$$

750 Moreover, using $\mathbb{E}[\xi_i \mid w_i] = 0$,

$$\begin{aligned} \mathbb{E}[\|\nabla L(w_i) + \Delta_i + \xi_i\|^2 \mid w_i] &= \|\nabla L(w_i) + \Delta_i\|^2 + \mathbb{E}[\|\xi_i\|^2 \mid w_i] \\ &\leq 2\beta^2 \|w_i - w^*\|^2 + 2B(p)^2 + \sigma^2. \end{aligned}$$

751 Combining these bounds yields

$$\begin{aligned} \mathbb{E}[\|w_{i+1} - w^*\|^2 \mid w_i] &\leq \left(1 - \frac{3\alpha\eta_i}{2} + 2\beta^2\eta_i^2\right) \|w_i - w^*\|^2 \\ &\quad + \frac{2\eta_i}{\alpha} B(p)^2 + \eta_i^2 \{2B(p)^2 + \sigma^2\}. \end{aligned}$$

752 Since $\eta_i \leq \alpha/(6\beta^2)$, we have

$$2\beta^2\eta_i^2 \leq \frac{\alpha\eta_i}{3},$$

753 and hence

$$1 - \frac{3\alpha\eta_i}{2} + 2\beta^2\eta_i^2 \leq 1 - \alpha\eta_i.$$

754 Taking expectations and absorbing constants gives

$$\delta_{i+1} \leq (1 - \alpha\eta_i)\delta_i + C\eta_i B(p)^2 + C\eta_i^2 \sigma^2,$$

755 where $C > 0$ depends only on α, β and the step-size constants.

756 It remains to solve this recursion. Define

$$\Lambda_i := \delta_i - KB(p)^2,$$

757 where $K > 0$ is chosen large enough such that, for all i ,

$$(1 - \alpha\eta_i)KB(p)^2 + C\eta_i B(p)^2 \leq KB(p)^2.$$

758 Equivalently, it is enough to take $K \geq C/\alpha$. Then

$$\Lambda_{i+1} \leq (1 - \alpha\eta_i)\Lambda_i + C\eta_i^2 \sigma^2.$$

759 With $\eta_i = c/(i + \gamma)$ and $\alpha c > 1$, the standard recursion bound gives

$$\Lambda_n = O(n^{-\alpha c}) + O\left(\frac{\sigma^2}{n}\right).$$

760 Because $\alpha c > 1$, the initialization term is $O(1/n)$. Therefore,

$$\delta_n = \mathbb{E}\|w_n - w^*\|^2 = O(B(p)^2) + O\left(\frac{\sigma^2}{n}\right).$$

761 If $\sigma^2 = O(1)$, this becomes

$$\mathbb{E}\|w_n - w^*\|^2 = O(B(p)^2) + O(1/n).$$

762 The $O(B(p)^2)$ term is the limiting neighborhood induced by the systematic gradient bias. $\square$

763 **Plug-in for plain imputed SGD.** Under independent hMCAR/sMAR, Proposition 1 gives $b(p) =$
764 $O(\|p\|)$. The variance of $\hat{g}^{(p)}$ is bounded by a constant under the standing L^2 assumptions. Hence

$$\mathbb{E}\|\bar{w}_n - w^*\|^2 = O(\|p\|^2) + O(1/n).$$

765 **Plug-in for Richardson-SGD.** Under independent hMCAR/sMAR, Corollary 2 gives $b(p) =$
766 $O(\|p\|^2)$. Using (12), σ^2 remains $O(1)$ in the moderate- C regime. Substituting in (32) yields

$$\mathbb{E}\|\bar{w}_n - w^*\|^2 = O(\|p\|^4) + O(1/n).$$

767 The analogous statement for the test loss follows by smoothness of L around w^* . The k -step variant
768 gives $b(p) = O(\|p\|^{k+1})$ and a missingness floor $O(\|p\|^{2(k+1)})$.

769 **Plug-in version with estimated mechanism.** Combining Proposition 3 and (32), the one-pass
770 bound becomes

$$\mathbb{E}\|\bar{w}_n - w^*\|^2 = O(\|p\|^4 + \delta_p^2 + \|p\|_\infty^2 \delta_q^2) + O(1/n).$$

771 Whenever δ_p, δ_q shrink at rate $o(\|p\|^2)$, the exact-mechanism rate is recovered.

772 **Multi-epoch behavior.** The above analysis only covers one pass, else the imputed gradients seen
773 on different epochs are not independent. The empirical study of Section 6 indicates that Richardson
774 remains effective in multi-epoch training; a formal multi-epoch analysis is left to future work.

775 G Implementation details

776 This appendix describes the experimental protocol used in Section 6 and in Appendix I below.
777 All experiments are run with stochastic gradient descent for 5 epochs, minibatch size 64, average
778 missingness level $\bar{p} = 0.20$, and first-order Richardson scale $C = 2$. Unless stated otherwise, all
779 reported curves are averaged over repeated runs with the same protocol across methods.

780 **Models.** We consider three generalized linear models: linear regression with Gaussian noise,
781 logistic regression for binary classification, and Poisson regression for count responses. All models
782 are trained with an ℓ_2 penalty. The regularization parameter is fixed to $\lambda = 10^{-3}$ for every model
783 family and dataset.

784 **Missingness mechanisms.** We evaluate three missingness mechanisms. The first is homogeneous
785 MCAR, denoted `mcAR`, where each entry is missing independently with the same probability p .
786 The second is heterogeneous MCAR, denoted `hetero_mcar`, where missingness probabilities are
787 generated from row and column multipliers and then calibrated to have average missingness $\bar{p}$.
788 Concretely, the unnormalized missingness scores are sampled uniformly in $[0, 1]$ across covariates
789 and rescaled so that their empirical mean equals 0.20.

790 The third mechanism is scalable MAR, denoted `smAR`. In this case, the oracle missingness intensity is

$$\lambda_j = p_j Q(U_j), \quad Q(u) = \sigma(1.6u - 0.3), \quad U = a_{1,j}X_1 + b_{2,j}X_2, \quad a_{1,\cdot}, b_{2,\cdot} \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(0, 1).$$

791 where σ is the logistic sigmoid. The coordinate-specific constants p_j are calibrated so that the average
792 missingness is $\bar{p} = 0.20$. This is the same scalable MAR mechanism as in Section 2.

793 **Methods compared.** We compare the complete-data baseline, plain imputation-based SGD, and
794 Richardson-corrected SGD. The complete-data baseline, denoted `No missing vals`, is trained
795 on the clean unmasked training data. The plain imputation baselines are zero imputation, mean
796 imputation, k -nearest-neighbor imputation, MICE, and MICE with random-forest base learners,
797 denoted respectively by `Zero`, `Mean`, `KNN`, `MICE`, and `MICE+RF`. The corresponding Richardson
798 variants are denoted `Rich.-Zero`, `Rich.-Mean`, `Rich.-KNN`, `Rich.-MICE`, and `Rich.-MICE+RF`.
799 All imputers are implemented using `scikit-learn`. The experiments are repeated 30 times with
800 different seeds, for the training of SGD methods, and averaged results, along their standard deviations,
801 are displayed.

E: all default parameters for imputation methods? A préciser.

802 **Metric.** The main metric is the parameter mean-squared error

$$\text{MSE}_w(t) = \frac{1}{d_w} \|\hat{w}_t - w^*\|_2^2,$$

803 where d_w is the parameter dimension, $\hat{w}_t$ is the SGD iterate after epoch t , and w^* is the complete-data
804 reference parameter described below. The metric is reported once per epoch for 5 epochs. For real
805 datasets, w^* denotes the minimizer of the complete-data penalized empirical risk, not a population
806 ground truth (see the paragraph *Reference parameter* below).

E: penalized? Comment ?

807 **Learning-rate calibration.** The optimization geometry varies substantially across model families
808 and datasets. To avoid confounding imputation effects with poorly tuned learning rates, we calibrate
809 the initial learning rate η_0 separately for each pair of model family and dataset.

810 For each pair, we first take the family-level default learning rate $\eta_0 = 10^{-2}$. We then evaluate the
811 logarithmic grid

$$\eta_0 \in \eta_0^{\text{def}} \cdot \left\{ \frac{1}{4}, \frac{1}{2}, 1, 2, 4 \right\}.$$

812 For every candidate, we run SGD, without missing data, on the standardized training fold using
813 the same number of epochs, minibatch size, and regularization parameter as in the missing-data
814 experiments. We select the learning rate that minimizes the final iteration parameter MSE,

$$\frac{1}{d_w} \|\hat{w}_T - w^*\|_2^2.$$

815 This calibration is performed without missingness and without imputation. The selected learning rate
816 is then fixed and reused for all imputation methods, Richardson variants, and missingness mechanisms
817 for that model–dataset pair. Thus, comparisons between MCAR, heterogeneous MCAR, and sMAR
818 within the same row use the same calibrated η_0 .

819 **Dataset budget and preprocessing.** Each dataset uses 2,000 training samples. Real datasets
820 with fewer observations are bootstrapped to this size when needed. Test sets contain 1000 samples.
821 Covariates are standardized columnwise on the training fold and the same transformation is applied
822 to the test fold.

823 The response variable is rescaled depending on the model family. For linear regression on real
824 datasets, the response is z-scored on the loaded sample. This keeps the scale of w^* comparable
825 across synthetic and real datasets; in particular, using raw elevation in the Coverttype regression task
826 produces parameters much larger than those in the synthetic linear experiments. Synthetic linear
827 responses are left unchanged, since the data-generating process already gives a comparable response
828 scale.

829 For logistic regression, the response is binary and no rescaling is applied. For Poisson regression,
830 real count responses are rescaled to have mean approximately 2 and then rounded to integer counts,
831 corresponding to a log-mean near 0.7.

832 **Reference parameter.** For synthetic datasets, w^* is the parameter used in the data-generating
833 process. The covariates are generated to be approximately centered and standardized, so the train-fold
834 standardization is nearly idempotent and the generating parameter remains the appropriate reference.
835 For real datasets, there is no closed-form ground-truth parameter. We therefore compute w^* by
836 L-BFGS-B optimization of the exact penalized empirical loss on the complete, standardized training
837 data, using the same regularization parameter $\lambda = 10^{-3}$ as in the SGD runs. This gives the complete-
838 data regularized minimizer of the observed sample loss and serves as the reference parameter for the
839 reported MSE.

E: via L2?

840 **Datasets.** The datasets used in the experiments are listed in Table 1. Synthetic datasets are generated
841 with Gaussian covariates. Real datasets are taken from standard `scikit-learn` or OpenML sources
842 and transformed as indicated. For the Bike Sharing Demand dataset, we use the hourly rental count
843 as the response and retain eight numeric features: year, month, hour, weekday, temperature, feeling
844 temperature, humidity, and windspeed.

Table 1: Datasets used in the experiments.

Family	#	Dataset	Source and preprocessing
Linear	1	Synth-A	Synthetic, 10D iid Gaussian
Linear	2	Synth-B	Synthetic, 15D AR-style covariance $\Sigma_{jk} = 0.9^{ j-k }$
Linear	3	Diabetes	Real, bootstrapped, z-scored response
Linear	4	Covertypes-reg	Real, z-scored elevation from 9 continuous features
Logistic	1	Synth-A	Synthetic, 10D iid Gaussian
Logistic	2	Breast cancer	Real, bootstrapped to 2,000 samples
Logistic	3	Covertypes	Real, class 1 versus all, 10 continuous features
Logistic	4	California housing	Real, response equal to price above median
Poisson	1	Synth-A	Synthetic, 10D iid Gaussian
Poisson	2	Synth-B	Synthetic, 8D iid Gaussian
Poisson	3	Bike sharing	Real, hourly rental count from numeric features

H Robustness to errors in the estimated missingness mechanism

The previous experiments kept p and q known as oracles. In practice, however, these quantities need to be estimated, and Proposition 3 provides an upper bound of the error induced by such estimations. In the following experiment, we test the impact of estimating p and q on Richardson-SGD, on top of imputation by zero, for logistic regression in the hMCAR setting. We perturb the estimated mechanism $(\hat{p}, \hat{q})$ by additive noise with magnitudes δ_p, δ_q and report the parameter MSE as a function of (δ_p, δ_q) . Table 2 shows the robustness of Richardson to plug-in estimation. With a reference parameter MSE of 2.647×10^{-2} for no missing data and 5.399×10^{-2} for plain imputation by zero, we see that even under high ratio mismatch, Richardson performs better than simple imputation. The only worse errors occur when $\delta_p = 0.3$ and $\delta_q \geq 0.2$, which we put in italics.

Table 2: MSE across δ_q and δ_p ; all entries are $\times 10^{-2}$.

$\delta_q \backslash \delta_p$	0.00	0.05	0.10	0.15	0.20	0.30
0.00	3.27	3.53	3.75	3.97	4.24	4.83
0.05	3.60	3.83	4.14	4.40	4.67	5.25
0.10	3.70	3.89	4.13	4.40	4.57	5.05
0.15	3.96	4.17	4.37	4.66	4.89	5.31
0.20	4.34	4.51	4.73	4.93	5.12	<i>5.49</i>
0.30	4.62	4.77	4.89	4.99	5.13	<i>5.40</i>

I Additional GLM experiments

We provide additional comparisons of SGD with several imputation rules, with and without Richardson, across linear (Gaussian), logistic, and Poisson regression on synthetic and real datasets. Each figure shows the parameter-MSE trajectory under SGD with and without Richardson on top of several imputation schemes; the bias formulas of App. D predict the per-model behavior. We organize the figures by GLM family $\times$ missingness mechanism. The key empirical observations are as follows.

Key empirical observations.

- Richardson is consistently effective on top of mean, MICE, MICE-RF, and k -NN imputation across the three GLMs and the three mechanisms.
- Although the theory only covers one-pass SGD, Richardson remains robust empirically in multi-epoch training.
- In several settings, the gains are most pronounced in the first epoch, in agreement with the one-pass theory of Section 5.5.

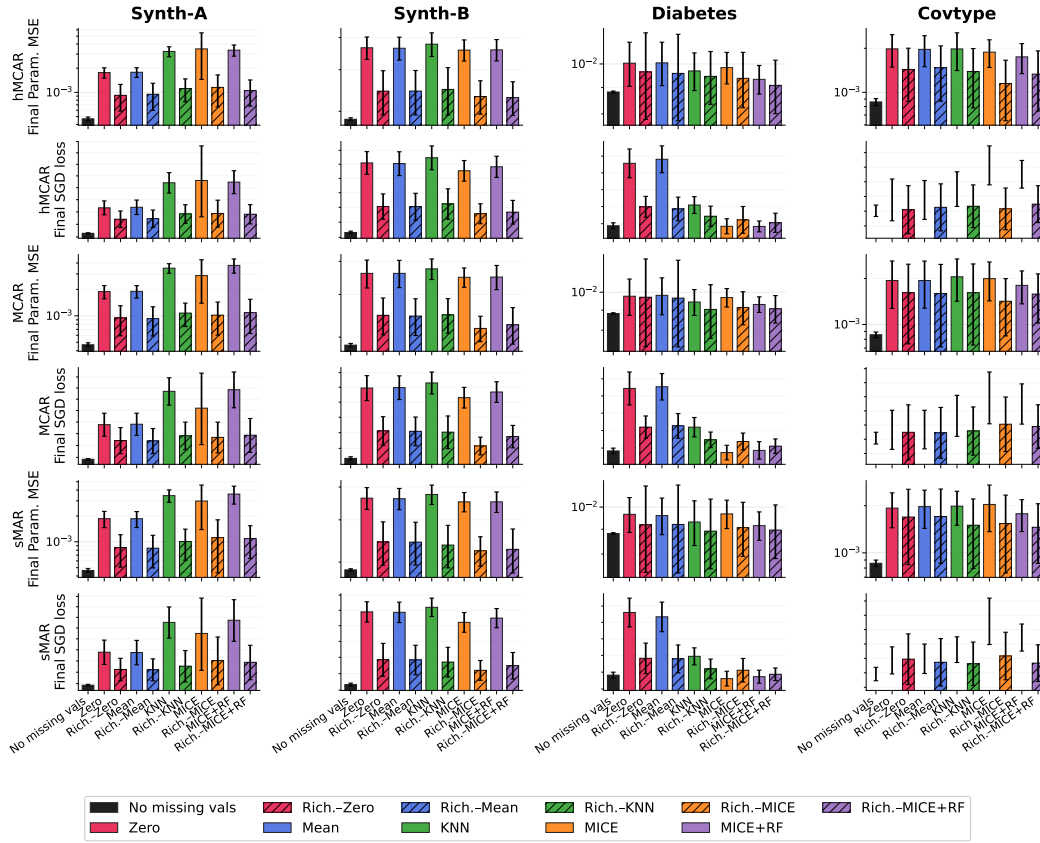
Linear regression — Final parameter error and final SGD loss (mean $\pm 1\sigma$ over 30 MC reps)

Figure 4: Final parameter MSE and test loss for linear regression, under MCAR, heterogeneous MCAR and sMAR mechanisms, on four different datasets.

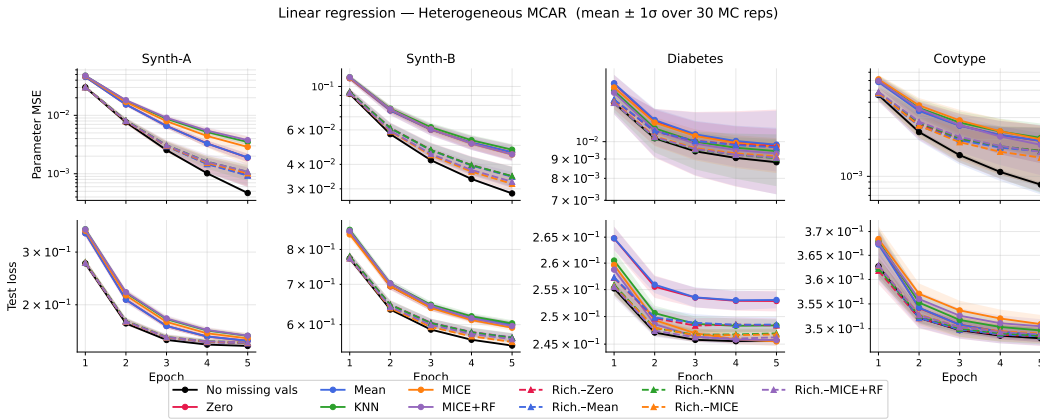
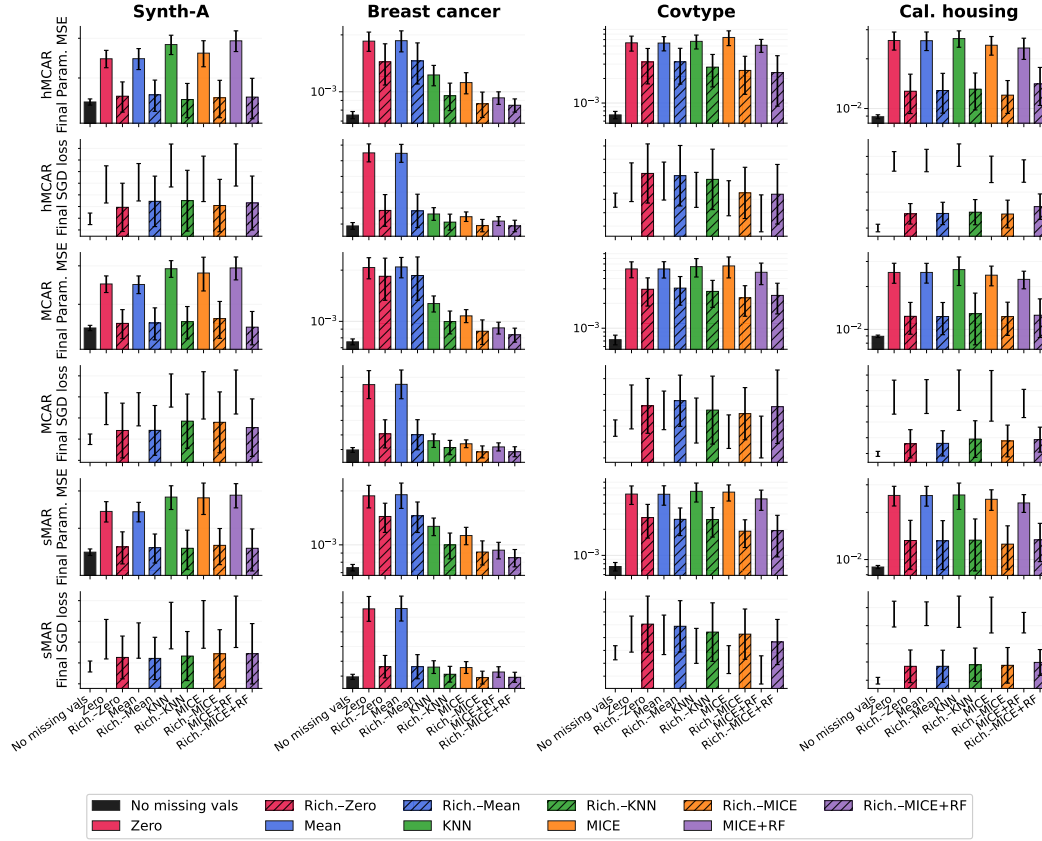
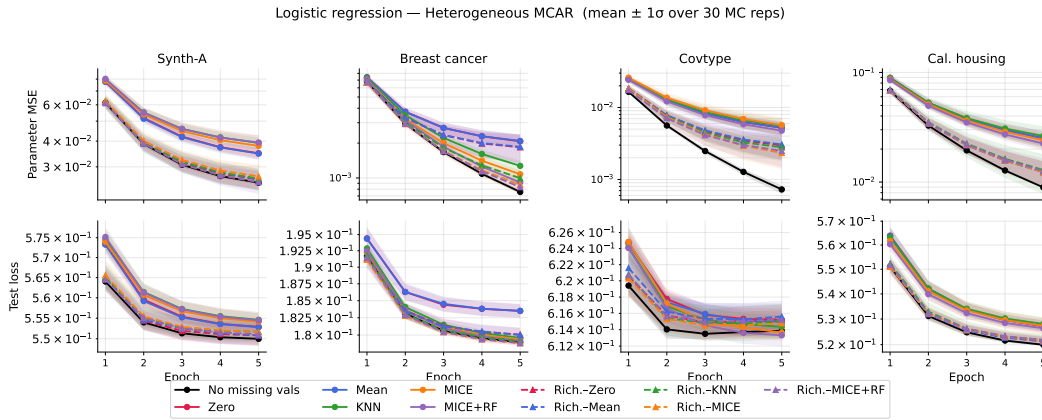


Figure 5: Convergence rate for parameter MSE and test loss for linear regression, under heterogeneous MCAR and sMAR mechanisms, on four different datasets.

Logistic regression — Final parameter error and final SGD loss (mean $\pm 1\sigma$ over 30 MC reps)**Figure 6: Final parameter MSE and test loss for logistic regression, under MCAR, heterogeneous MCAR and sMAR mechanisms, on four different datasets.****Figure 7: Convergence rate for parameter MSE and test loss for logistic regression, under heterogeneous MCAR and sMAR mechanisms, on four different datasets.**

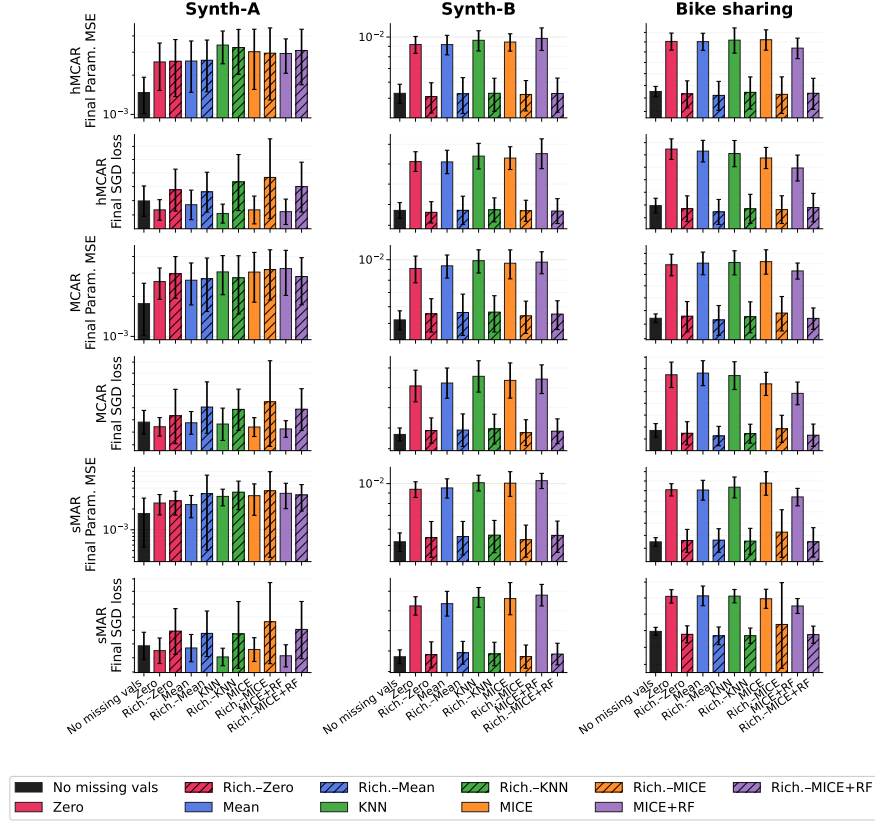
Poisson regression — Final parameter error and final SGD loss (mean $\pm 1\sigma$ over 30 MC reps)

Figure 8: Final parameter MSE and test loss for Poisson regression, under MCAR, heterogeneous MCAR and sMAR mechanisms, on four different datasets.

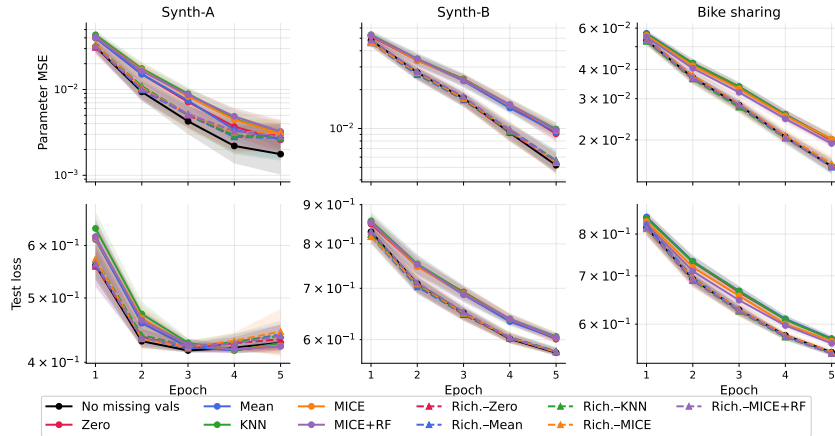
Poisson regression — Heterogeneous MCAR (mean $\pm 1\sigma$ over 30 MC reps)

Figure 9: Convergence rate for parameter MSE and test loss for Poisson regression, under heterogeneous MCAR and sMAR mechanisms, on four different datasets.

871 J Comparison with other schemes

872 The experiments above show that Richardson-SGD improves performance across the three GLMs
873 considered and across all imputation techniques tested. This is the regime for which the method is
874 primarily intended: when the learner has access to an imputation pipeline, but does not want to impose
875 a model-specific correction or strong structural assumptions on the covariate distribution. In this
876 sense, Richardson-SGD is best viewed as a model-agnostic and distribution-agnostic debiasing layer
877 on top of imputation, rather than as a competitor to specialized estimators designed for a particular
878 statistical model. Consequently, the most direct comparison is with the same imputation pipeline
879 used without Richardson.

880 For completeness, we nevertheless include a more specialized benchmark in the linear-regression
881 setting. In this case, tailored alternatives are available: debiased SGD under hMCAR [32], and
882 SAEM-type methods under parametric assumptions on the covariate distribution [11]. These methods
883 are designed specifically for this setting, so Richardson-SGD is not expected to dominate them. The
884 point of the comparison is instead to test whether a generic Richardson correction remains competitive
885 even in a regime where model-specific methods have an intrinsic advantage.

886 The results support this interpretation. Richardson-SGD performs close to debiased SGD and im-
887 proves over SAEM, especially on non-synthetic datasets where the parametric assumptions underlying
888 SAEM are less well matched to the data. The main failure case occurs for Richardson combined with
889 MICE in the uncorrelated Gaussian-covariate setting. This behavior is expected: when covariates are
890 independent, MICE has essentially no cross-feature signal to exploit and may fit noise, making it a
891 poor base imputer. In such cases, Richardson inherits the limitations of the underlying imputation
892 procedure.

Table 3: Parameter MSE and runtime results. Values are reported as mean $\pm$ standard deviation. SAEM is run only once, with a runtime of 5 s for Synth-A and Synth-B, and 100 s otherwise.

Dataset	Method	PMSE	Time (s)
Synth-A	No missing data	$5.51 \times 10^{-5} \pm 1.10 \times 10^{-6}$	0.019
Synth-A	Zero-Impute	$4.11 \times 10^{-3} \pm 1.00 \times 10^{-3}$	0.021
Synth-A	Rich. – Zero	$2.36 \times 10^{-4} \pm 3.70 \times 10^{-5}$	0.028
Synth-A	Rich. – MICE	$1.56 \times 10^{-2} \pm 9.60 \times 10^{-3}$	0.036
Synth-A	Debiased SGD (Sportisse et al.)	$2.99 \times 10^{-4} \pm 1.10 \times 10^{-4}$	0.057
Synth-A	SAEM	$6.00 \times 10^{-4} \pm 0$	5.00
Synth-B	No missing data	$2.39 \times 10^{-3} \pm 1.50 \times 10^{-5}$	0.029
Synth-B	Zero-Impute	$4.95 \times 10^{-1} \pm 3.22 \times 10^{-2}$	0.031
Synth-B	Rich. – Zero	$4.97 \times 10^{-2} \pm 1.80 \times 10^{-3}$	0.055
Synth-B	Rich. – MICE	$3.42 \times 10^{-2} \pm 4.00 \times 10^{-3}$	0.057
Synth-B	Debiased SGD (Sportisse et al.)	$4.64 \times 10^{-3} \pm 2.00 \times 10^{-3}$	0.073
Synth-B	SAEM	$2.33 \times 10^{-2} \pm 0$	5.00
Covtype	No missing data	$1.94 \times 10^{-2} \pm 0$	0.028
Covtype	Zero-Impute	$8.04 \times 10^{-2} \pm 5.05 \times 10^{-2}$	0.033
Covtype	Rich. – Zero	$2.60 \times 10^{-2} \pm 3.55 \times 10^{-3}$	0.055
Covtype	Rich. – MICE	$2.37 \times 10^{-2} \pm 1.50 \times 10^{-3}$	0.057
Covtype	Debiased SGD (Sportisse et al.)	$1.99 \times 10^{-2} \pm 5.50 \times 10^{-3}$	0.079
Covtype	SAEM	$2.58 \times 10^{-1} \pm 0$	100.00
Cal. housing	No missing data	$2.62 \times 10^{-2} \pm 5.00 \times 10^{-5}$	0.026
Cal. housing	Zero-Impute	$1.02 \times 10^{-1} \pm 9.76 \times 10^{-3}$	0.021
Cal. housing	Rich. – Zero	$7.15 \times 10^{-2} \pm 1.23 \times 10^{-2}$	0.051
Cal. housing	Rich. – MICE	$5.10 \times 10^{-2} \pm 1.49 \times 10^{-2}$	0.049
Cal. housing	Debiased SGD (Sportisse et al.)	$3.16 \times 10^{-2} \pm 9.55 \times 10^{-3}$	0.065
Cal. housing	SAEM	$2.70 \times 10^{-1} \pm 0$	100.00

E: Tu pourrais faire la même chose pour la régression logistique car dans ce cas, on veut juste se comparer à SAEM, pour lequel on fait probablement mieux sur données réelles. Cela peut être sur un jeu de données seulement.

E: on n'arrive jamais à faire mieux que Debiased SGD, que ce soit en temps de calcul ou en performance? On peut essayer de réfléchir à des situations favorables

E: mettre en gras dans chaque dataset l'erreur la plus basse et toutes celles situées à moins d'un écart-type de distance de celle-ci?

896 K Robustness to misspecification of the missingness mechanism

897 This appendix tests Richardson-SGD under misspecification of the thinning mechanism. Missing
 898 values are generated under the sMAR mechanism of Appendix G, where missingness depends on X_1
 899 and X_2 , but Richardson thinning uses the hMCAR approximation p_j instead of the true conditional
 900 probabilities $p_j q_j(V)$. We run linear regression on the four datasets in Table 1, a setting where the
 901 zero-imputation gradient bias is a polynomial of degree at most two in the missingness probabilities;
 902 see Appendix E. We compare zero imputation and MICE, with and without first-order Richardson
 903 correction, over 10 runs. Figure 10 shows the parameter-MSE and test-loss trajectories, and Table 4
 904 reports final values.

905 Richardson remains robust to this misspecification. With zero imputation, Richardson improves
 906 over plain zero imputation on all datasets, often nearly matching the complete-data baseline. With
 907 MICE, Richardson improves performance on Synth-A, Synth-B, and Diabetes. The main exception
 908 is California Housing, where Richardson-MICE becomes unstable near the last epoch; the plotted
 909 variance is capped for readability. This instability is consistent with the variance amplification of
 910 Richardson extrapolation discussed in Section 5.3, and may also reflect an imperfect learning-rate
 911 choice.

912 Overall, treating sMAR data as hMCAR does not eliminate the benefit of Richardson-SGD in these
 913 experiments, especially with zero imputation. However, the California Housing-MICE case shows
 914 that misspecification can interact with the imputation rule and optimization dynamics.

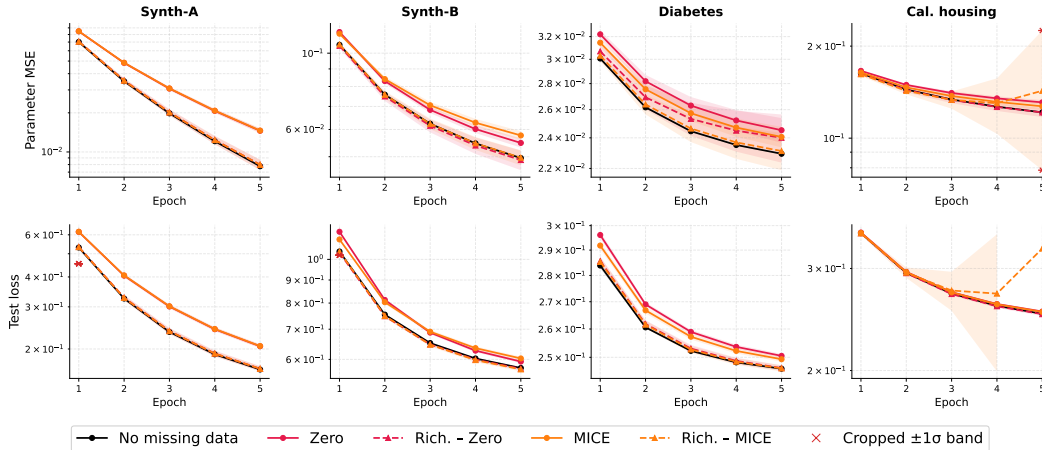


Figure 10: Richardson-SGD under misspecification of the missingness mechanism. Missing values are sMAR, but Richardson-SGD is computed using an hMCAR approximation based only on marginal missingness probabilities.

Table 4: Final parameter MSE and test loss for Richardson-SGD under misspecification of the missingness mechanism. The true mechanism is sMAR, while Richardson thinning uses an hMCAR approximation. Test losses are reported in units of 10^{-1} .

Method	Synth-A		Synth-B		Diabetes		Cal. housing	
	PMSE	Loss	PMSE	Loss	PMSE	Loss	PMSE	Loss
No missing data	$7.77 \pm 0.045 \cdot 10^{-3}$	1.64	$4.96 \pm 0.016 \cdot 10^{-2}$	5.74	$2.29 \pm 0.0037 \cdot 10^{-2}$	2.46	$1.21 \pm 0.00099 \cdot 10^{-1}$	2.51
Zero	$1.45 \pm 0.063 \cdot 10^{-2}$	2.05	$5.48 \pm 0.17 \cdot 10^{-2}$	5.93	$2.45 \pm 0.086 \cdot 10^{-2}$	2.51	$1.31 \pm 0.024 \cdot 10^{-1}$	2.53
Rich.-Zero	$7.99 \pm 0.75 \cdot 10^{-3}$	1.65	$4.89 \pm 0.32 \cdot 10^{-2}$	5.72	$2.40 \pm 0.16 \cdot 10^{-2}$	2.46	$1.21 \pm 0.041 \cdot 10^{-1}$	2.50
MICE	$1.46 \pm 0.061 \cdot 10^{-2}$	2.06	$5.77 \pm 0.17 \cdot 10^{-2}$	6.03	$2.41 \pm 0.070 \cdot 10^{-2}$	2.49	$1.27 \pm 0.035 \cdot 10^{-1}$	2.52
Rich.-MICE	$7.97 \pm 0.71 \cdot 10^{-3}$	1.65	$4.97 \pm 0.25 \cdot 10^{-2}$	5.70	$2.31 \pm 0.12 \cdot 10^{-2}$	2.46	$1.43 \pm 1.20 \cdot 10^{-1}$	3.25

L Why the two missingness scales must share the same imputation

Richardson correction compares two gradients evaluated at missingness scales p and Cp . For the linear term to cancel, these two gradients must be generated by the *same imputation operator*. In particular, entries that are missing at both scales must receive the same imputed value. This is why, in Section 4, we impute only once at the higher missingness scale Cp , and then restore the entries that were artificially hidden to obtain the lower-scale covariate.

We formalize this point. Let $\mathcal{I}$ be a data-independent imputation rule and define

$$\hat{g}_{\mathcal{I}}^{(p)}(w) := g(w; \tilde{X}_{\mathcal{I}}^{(p)}, Y), \quad \mathcal{B}_{\mathcal{I}}(w, p) := \mathbb{E}[\hat{g}_{\mathcal{I}}^{(p)}(w)] - \nabla L(w).$$

By Proposition 1,

$$\mathcal{B}_{\mathcal{I}}(w, p) = \mathcal{A}_{\mathcal{I}}(w)p + \mathcal{R}_{\mathcal{I}}(w, p), \quad \mathcal{R}_{\mathcal{I}}(w, p) = O(\|p\|^2)$$

under independent hMCAR or independent sMAR. The first-order operator $\mathcal{A}_{\mathcal{I}}(w)$ depends on the imputation rule, since its j -th column is the expected gradient gap created by declaring coordinate j missing.

If Richardson is applied with two possibly different imputation rules $\mathcal{I}_0$ and $\mathcal{I}_1$ at scales p and Cp , respectively, then

$$\hat{g}_{C, \mathcal{I}_0, \mathcal{I}_1}^R(w) := \frac{C \hat{g}_{\mathcal{I}_0}^{(p)}(w) - \hat{g}_{\mathcal{I}_1}^{(Cp)}(w)}{C - 1}.$$

Its bias is

$$\begin{aligned} \mathbb{E}[\hat{g}_{C, \mathcal{I}_0, \mathcal{I}_1}^R(w)] - \nabla L(w) &= \frac{C \mathcal{B}_{\mathcal{I}_0}(w, p) - \mathcal{B}_{\mathcal{I}_1}(w, Cp)}{C - 1} \\ &= \frac{C}{C - 1} (\mathcal{A}_{\mathcal{I}_0}(w) - \mathcal{A}_{\mathcal{I}_1}(w))p + O(\|p\|^2). \end{aligned}$$

Thus the $O(\|p\|)$ term cancels only if

$$\mathcal{A}_{\mathcal{I}_0}(w) = \mathcal{A}_{\mathcal{I}_1}(w).$$

This condition is automatic when the two gradients are constructed from the same higher-scale imputation, as in Algorithm 10: common missing entries have identical imputed values at both scales, and the only difference between $\tilde{X}^{(p)}$ and $\tilde{X}^{(Cp)}$ comes from the entries artificially hidden by the thinning step.

By contrast, if one independently runs two stochastic imputers, for example two separate MICE chains at scales p and Cp , then common missing entries may receive different imputations. The corresponding first-order operators need not coincide, so Richardson may leave an even bigger uncancelled $O(\|p\|)$ bias and can amplify stochastic imputation noise through the factor $(C - 1)^{-1}$.

Figure 11 illustrates this effect on a linear-regression experiment on California Housing. The linked construction, which imputes once at scale Cp , $C = 1.5$, and restores artificially hidden entries, remains stable and improves over plain MICE. The unlinked construction, which runs independent MICE imputations at the two scales, loses the first-order cancellation and becomes unstable.

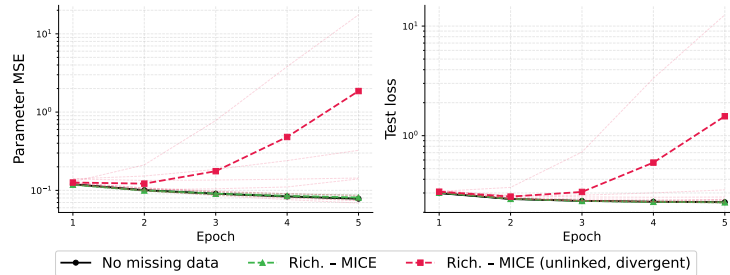


Figure 11: Linked versus unlinked Richardson–MICE on California Housing linear regression. The linked version uses one imputation at scale Cp and restores the artificially hidden entries to obtain the scale- p covariate. The unlinked version runs two separate MICE imputations at scales p and Cp . Only the linked construction preserves the common imputed values required for first-order Richardson cancellation.

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